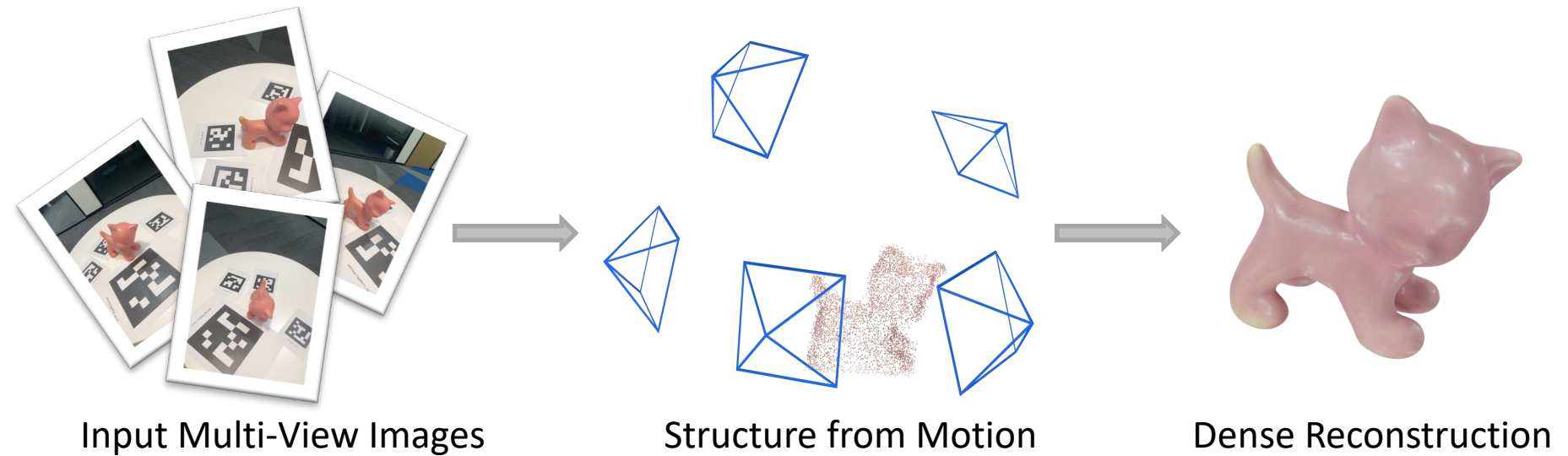


Introduction to Computer Vision



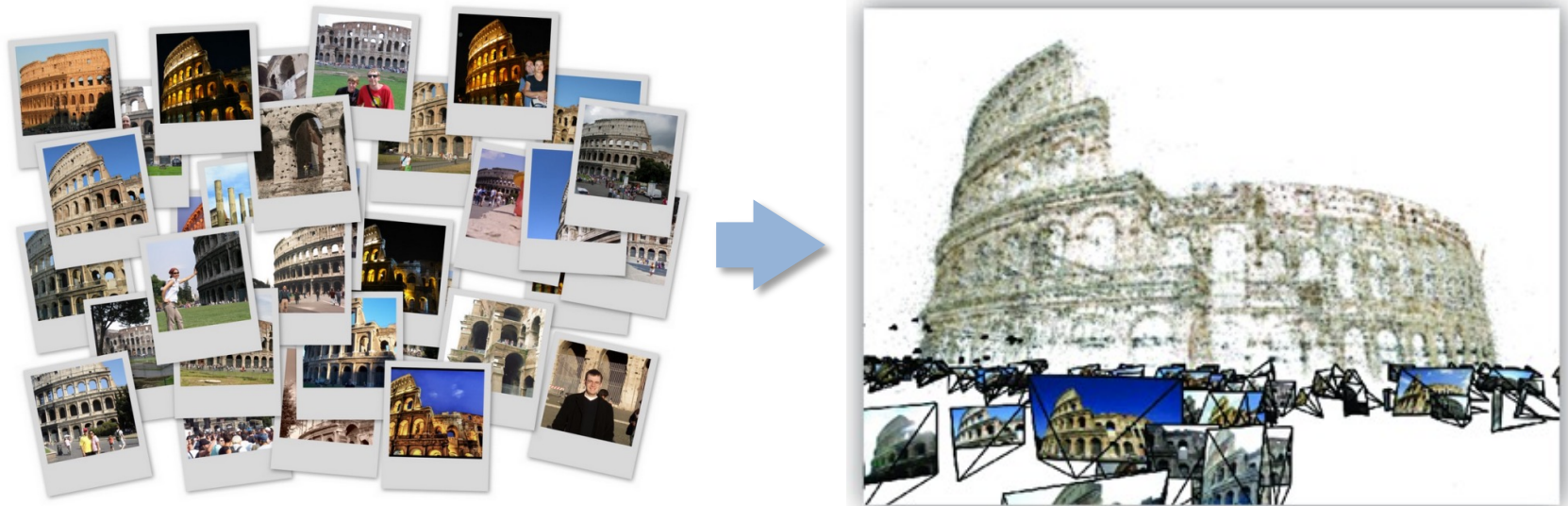
Lec.07: Structure from Motion
2022.10.31

Final Project



Structure from Motion (SfM)

Recover **camera poses** and **3D structure** of a scene from its images

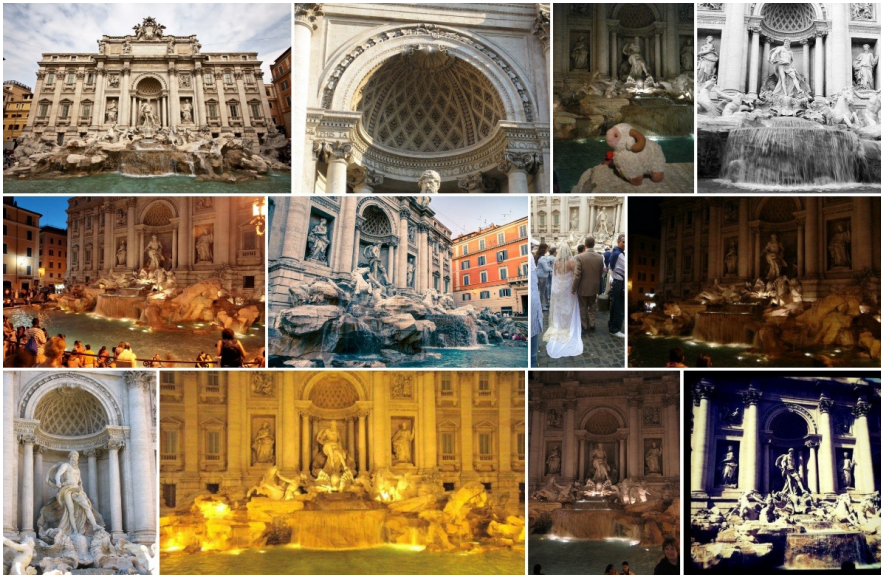


Structure from Motion (SfM)

Recover **camera poses** and **3D structure** of a scene from its images



SfM Extension: Multi-view Stereo



SfM Extension: Visual Localization



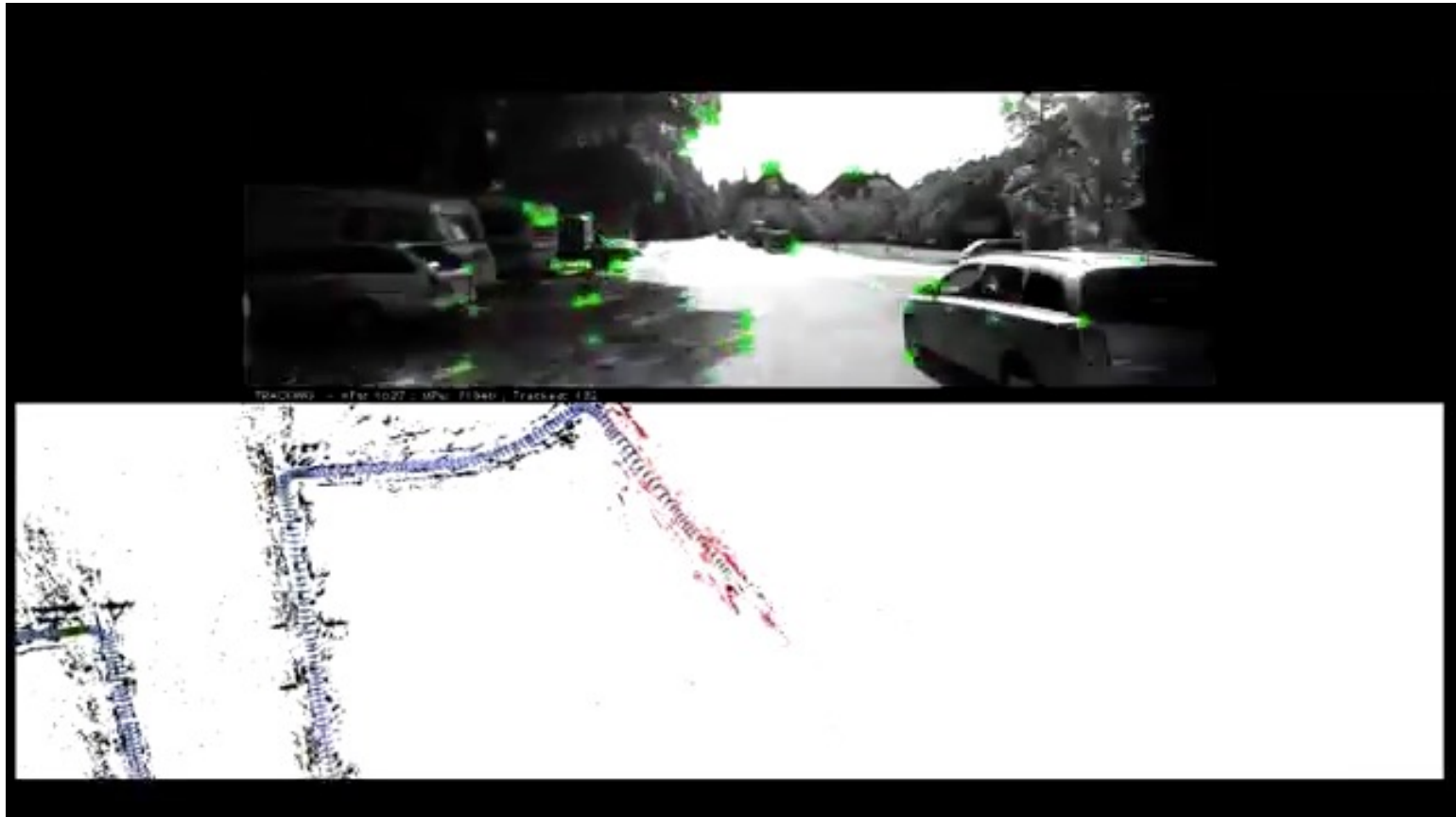
Known 3D Scene



Picture

Where was the picture taken?

SfM Extension: SLAM



Simultaneous localization and mapping

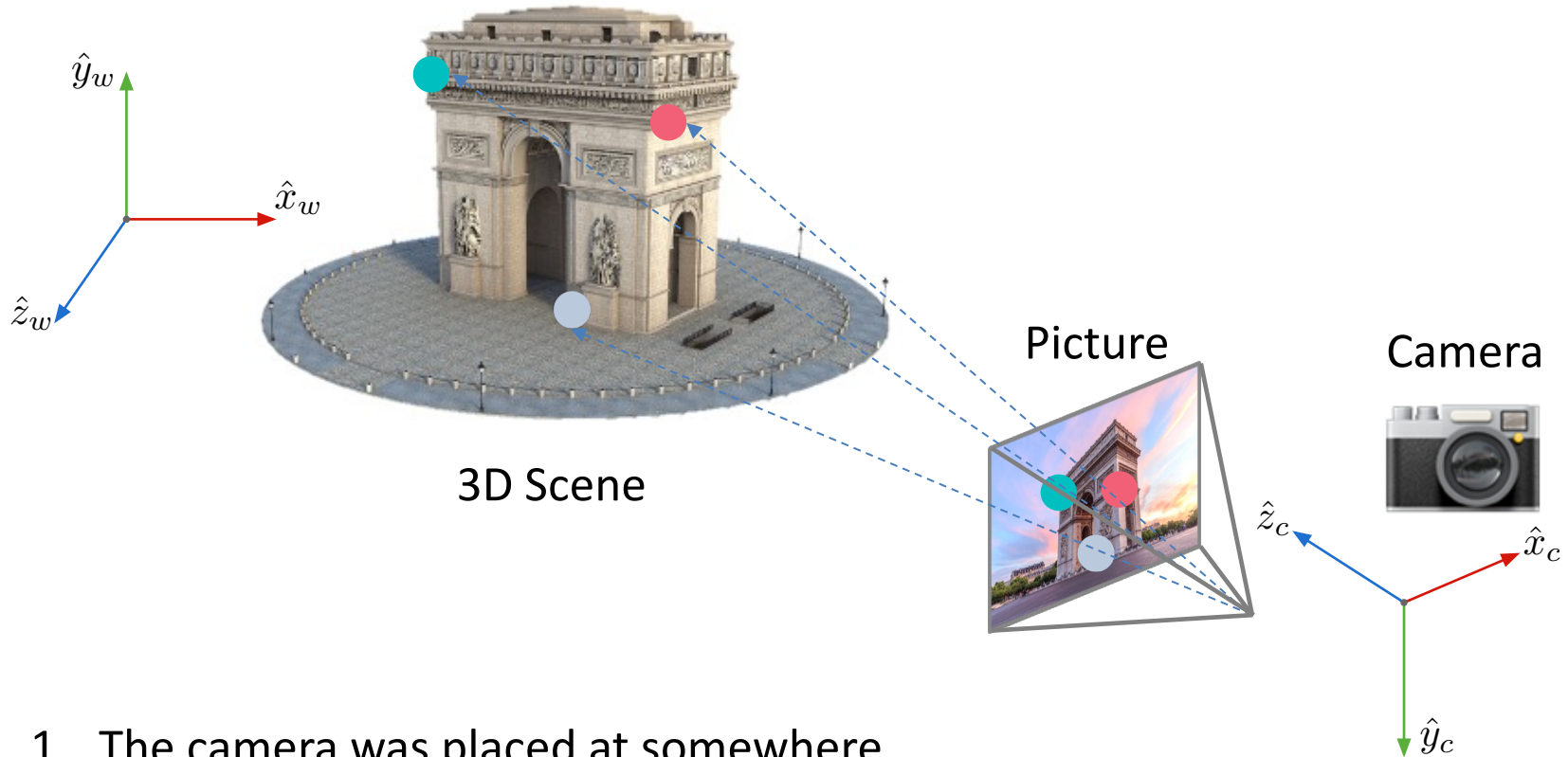
Problems to be noticed

1. How the camera maps the 3D points in the world onto its image plane. (**camera model**)
2. How to compute the position and orientation of the camera w.r.t. the world coordinate frame. (**camera calibration and pose estimation**)
3. How to reconstruct the unknown 3D structure from images (**structure from motion**)

Today

- 1. Camera model**
2. Camera calibration
3. Structure from motion

Image Formation



1. The camera was placed at somewhere.
2. The shutter was pressed.
3. A picture was created by photon-to-electron conversion.

1. Coordinate
Transformation



2. Perspective
Projection



3. Image Plane to
Image Sensor Mapping

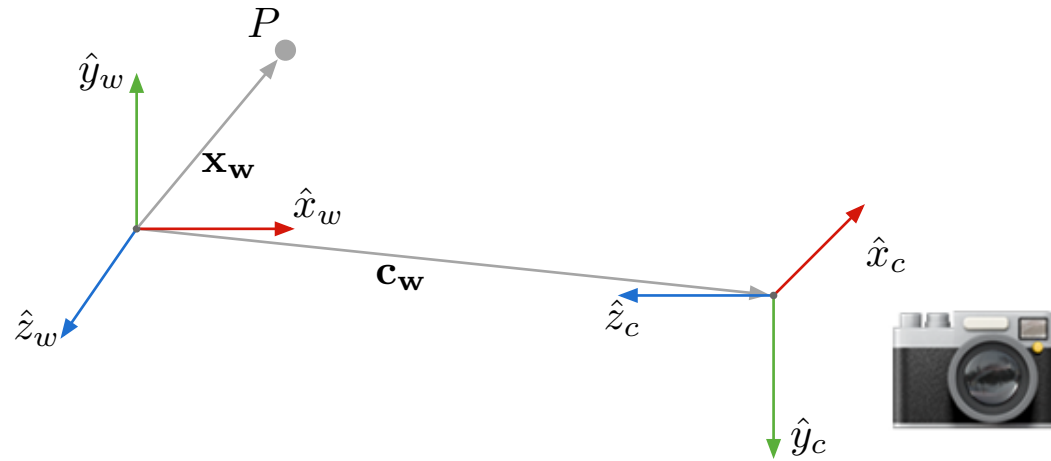
Image Formation



Extrinsic Matrix

Intrinsic Matrix

Coordinate Transformation

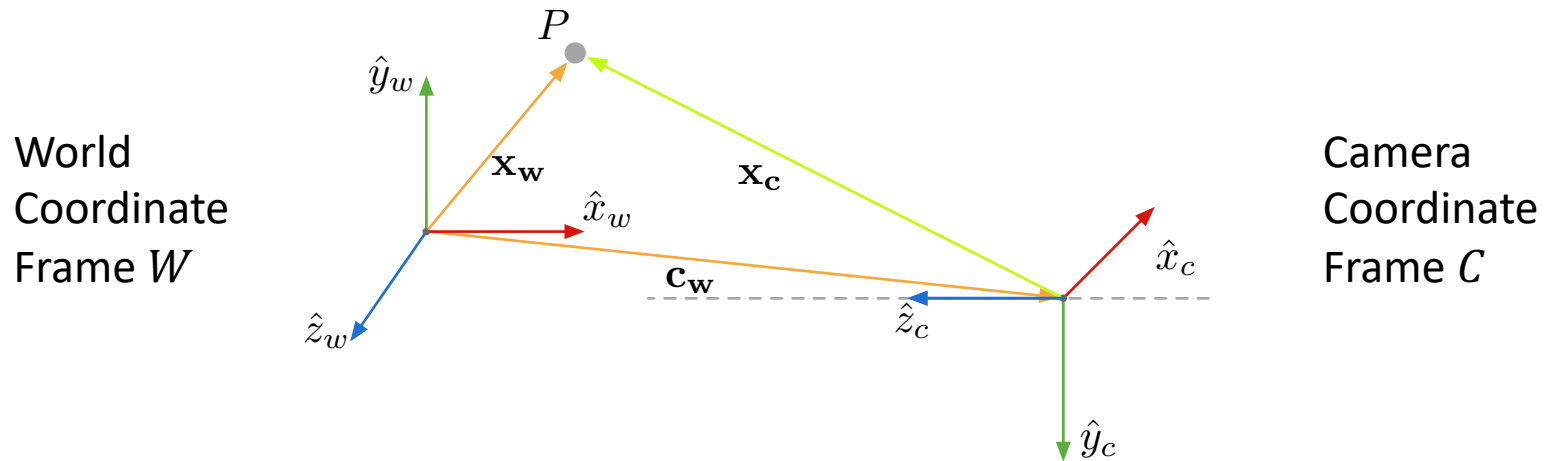


World Coordinates

Camera Coordinates

$$\mathbf{x}_w = \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} \quad \longrightarrow \quad \mathbf{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix}$$

Extrinsic Parameters



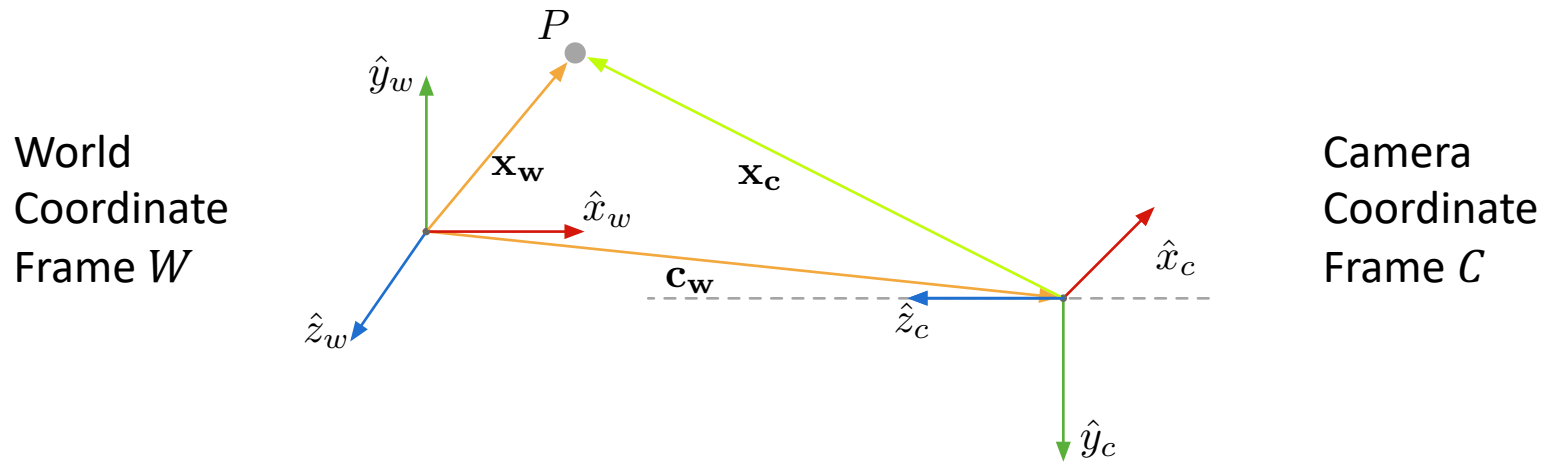
Position $\mathbf{c}_w$ and **Orientation** R of the camera in the world coordinate frame W are the camera's **Extrinsic Parameters**.

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

- Row 1: Direction of $\hat{x}_c$ in world coordinate frame
- Row 2: Direction of $\hat{y}_c$ in world coordinate frame
- Row 3: Direction of $\hat{z}_c$ in world coordinate frame

Rotation matrix R is orthonormal

World-to-Camera Transformation



Given the **extrinsic parameters** ($R, \mathbf{c}_w$) of the camera, the camera-centric location of the point P is:

$$\mathbf{x}_c = R(\mathbf{x}_w - \mathbf{c}_w) = R\mathbf{x}_w - R\mathbf{c}_w = R\mathbf{x}_w + \mathbf{t}$$

$$\mathbf{t} = -R\mathbf{c}_w$$

$$\mathbf{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

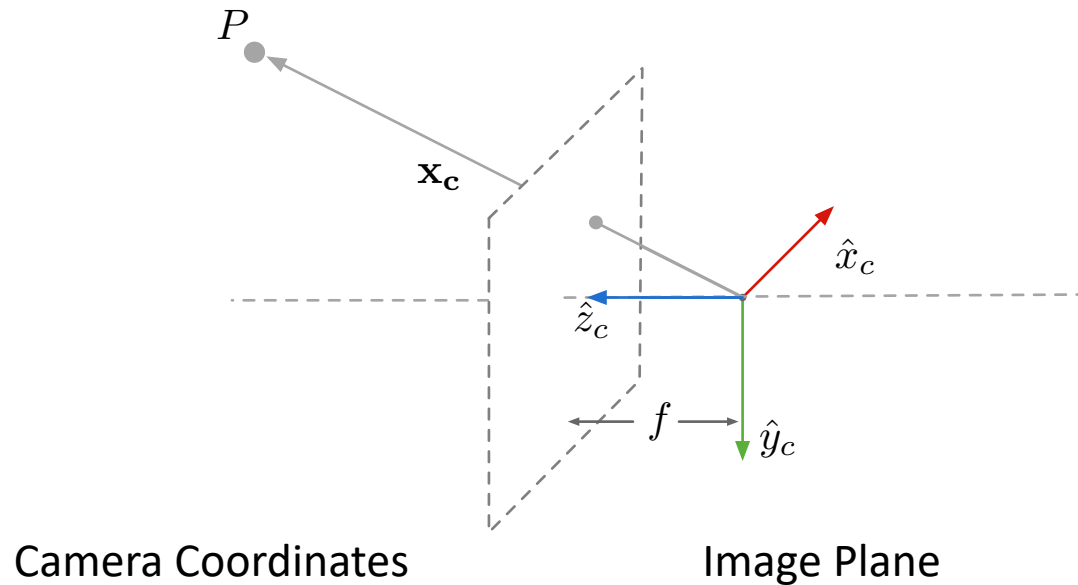
Extrinsic Matrix

Rewriting using homogenous coordinates:

$$\tilde{\mathbf{x}}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

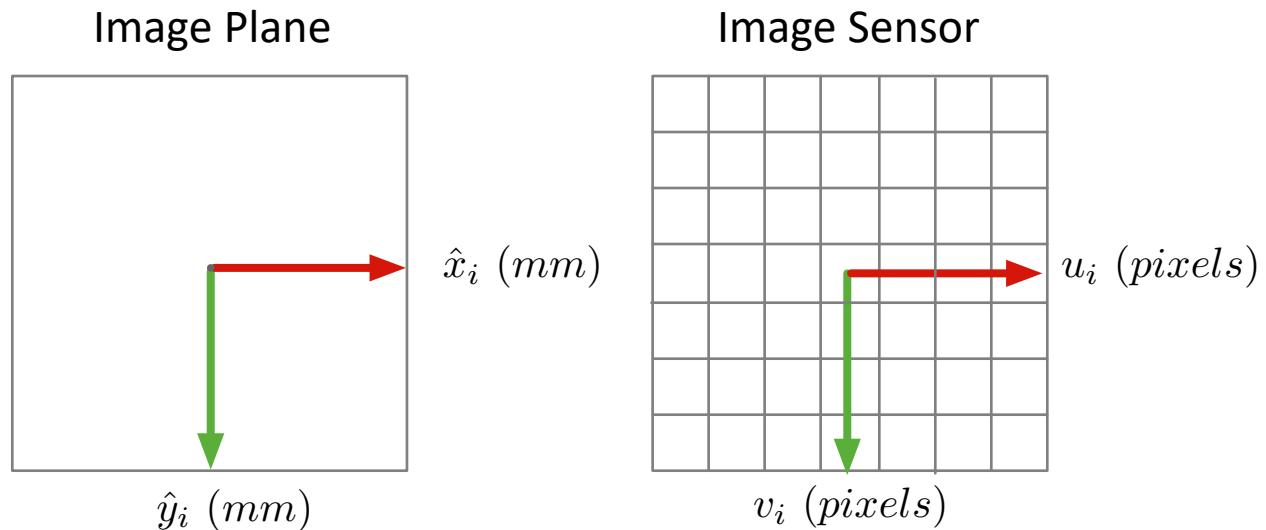
Extrinsic Matrix: $M_{ext} = \begin{bmatrix} R_{3 \times 3} & \mathbf{t} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Perspective Projection



$$\mathbf{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} \quad \longrightarrow \quad \mathbf{x}_i = \mathbf{f} \cdot \begin{bmatrix} x_c/z_c \\ y_c/z_c \\ 1 \end{bmatrix}$$

Image Plane to Image Sensor Mapping



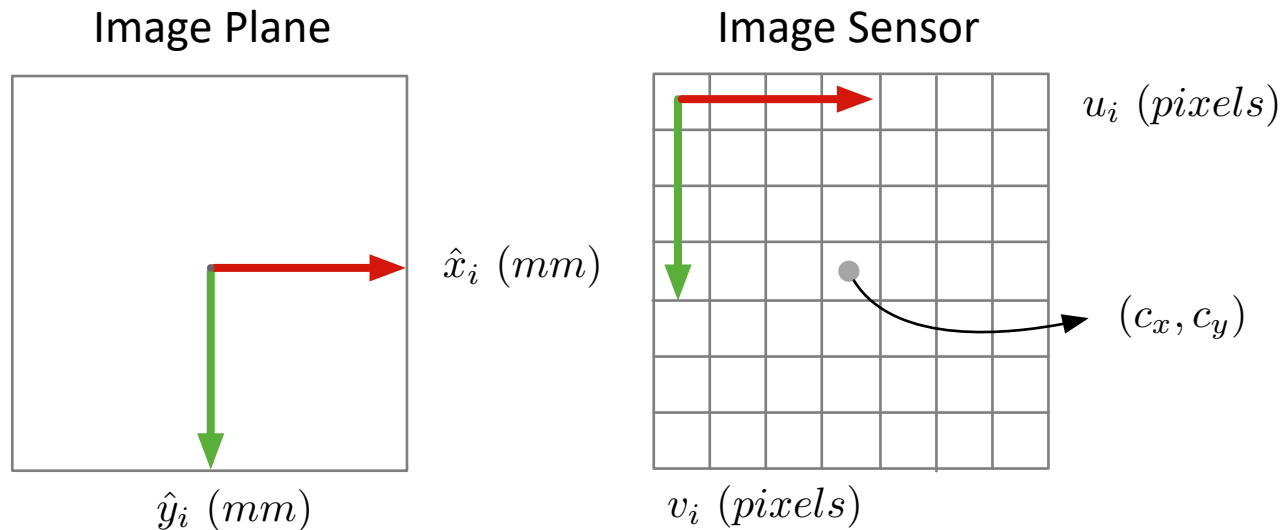
Pixels may be rectangular.

If m_x and m_y are the pixel densities (pixels/mm) in x and y directions, respectively, then pixel coordinates are:

$$u = m_x x_i = m_x f \frac{x_c}{z_c}$$

$$v = m_y y_i = m_y f \frac{y_c}{z_c}$$

Image Plane to Image Sensor Mapping

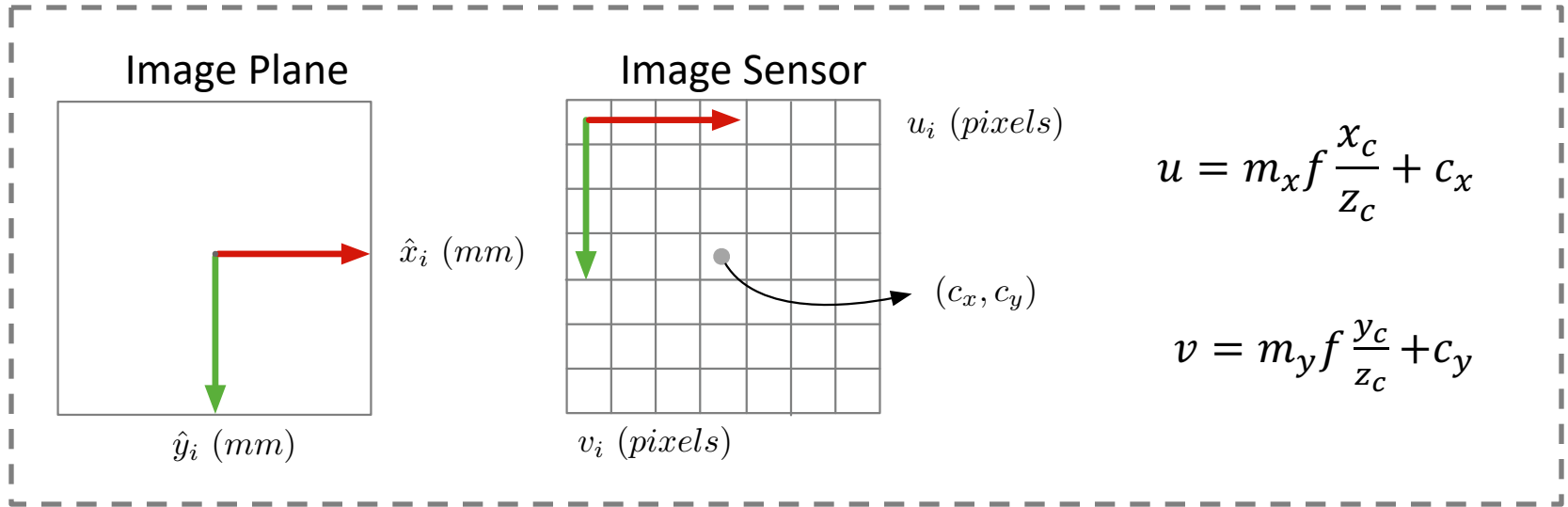


We usually treat the top-left corner of the image sensor as its origin (easier for indexing). If pixel (c_x, c_y) is the Principle Point where the optical axis pierces the sensor, then:

$$u = m_x f \frac{x_c}{z_c} + c_x$$

$$v = m_y f \frac{y_c}{z_c} + c_y$$

Intrinsic Matrix



In homogenous coordinates:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$

$$\begin{aligned} f_x &= m_x f \\ f_y &= m_y f \end{aligned}$$

Projection Matrix P

Camera to Pixel

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$

World to Camera

$$\begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

$$\tilde{\mathbf{u}} = \mathbf{M}_{int} \tilde{\mathbf{x}}_c$$

$$\tilde{\mathbf{x}}_c = \mathbf{M}_{ext} \tilde{\mathbf{x}}_w$$

Combining the above two equations, we get the full **projection matrix P**:

$$\tilde{\mathbf{u}} = \mathbf{M}_{int} \mathbf{M}_{ext} \tilde{\mathbf{x}}_w = \mathbf{P} \tilde{\mathbf{x}}_w$$

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

How to get those matrices in practice?

Today

1. Camera model
- 2. Camera calibration**
3. Structure from motion

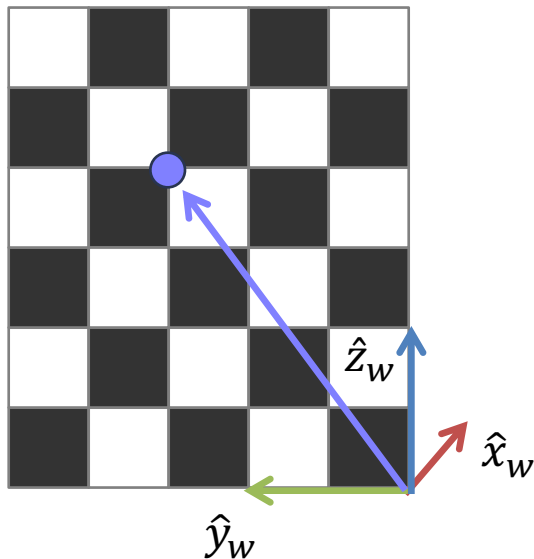
Camera Calibration

How to find the intrinsic and extrinsic parameters?

$$\tilde{\mathbf{u}} = M_{int} M_{ext} \tilde{\mathbf{x}}_w$$

Camera Calibration Procedure

Step 1: Capture an image of an object with known geometry, e.g. a calibration board

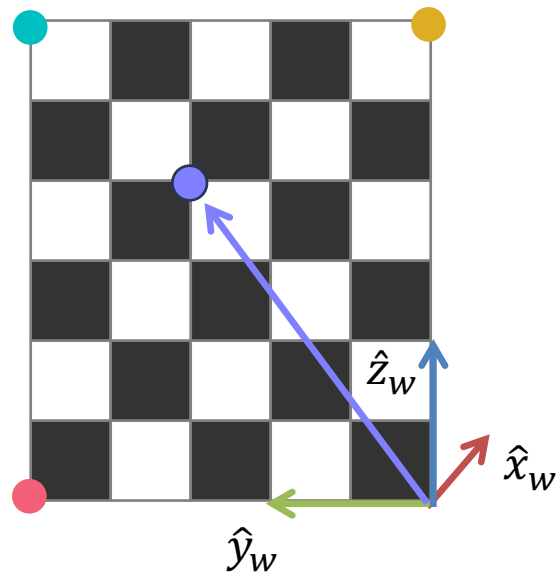


Object of Known Geometry

$$\bullet \quad \mathbf{x}_w = \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = \begin{bmatrix} 0 \\ 15 \\ 20 \end{bmatrix} \text{ cm}$$

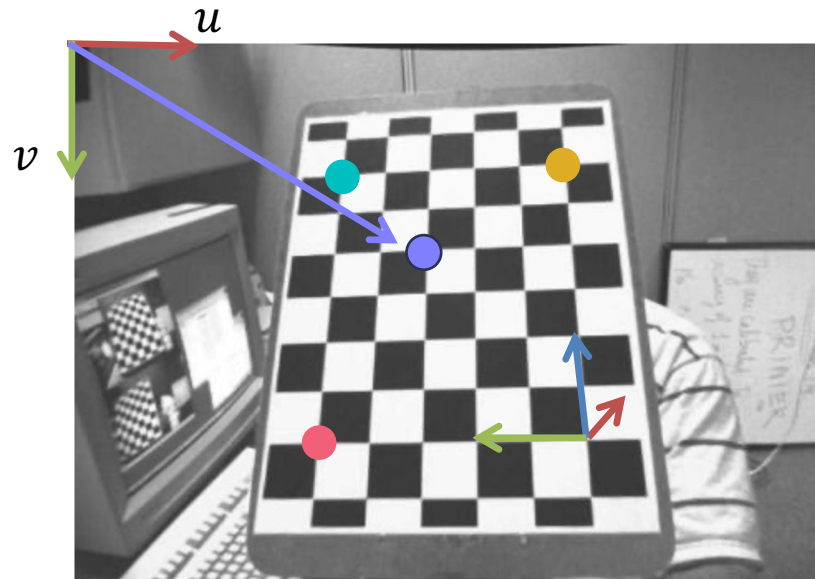
Camera Calibration Procedure

Step 2: Identify correspondences between 3D scene points and image points.



Object of Known Geometry

$$\bullet \quad \mathbf{x}_w = \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = \begin{bmatrix} 0 \\ 15 \\ 20 \end{bmatrix} \text{ cm}$$



Captured Image

$$\bullet \quad \mathbf{u}_i = \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 115 \\ 56 \end{bmatrix} \text{ pixels}$$

Camera Calibration Procedure

Step 3: For each corresponding point i in scene and image:

$$\underbrace{\begin{bmatrix} u^{(i)} \\ v^{(i)} \\ 1 \end{bmatrix}}_{\text{Known}} \equiv \underbrace{\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix}}_{\text{Unknown}} \underbrace{\begin{bmatrix} x_w^{(i)} \\ y_w^{(i)} \\ z_w^{(i)} \\ 1 \end{bmatrix}}_{\text{Known}}$$

Expanding the matrix as linear equations:

$$u^{(i)} = \frac{p_{11}x_w^{(i)} + p_{12}y_w^{(i)} + p_{13}z_w^{(i)} + p_{14}}{p_{31}x_w^{(i)} + p_{32}y_w^{(i)} + p_{33}z_w^{(i)} + p_{34}}$$
$$v^{(i)} = \frac{p_{21}x_w^{(i)} + p_{22}y_w^{(i)} + p_{23}z_w^{(i)} + p_{24}}{p_{31}x_w^{(i)} + p_{32}y_w^{(i)} + p_{33}z_w^{(i)} + p_{34}}$$

Camera Calibration Procedure

Step 4: Rearranging the terms

$$\underbrace{\begin{bmatrix} x_w^{(1)} & y_w^{(1)} & z_w^{(1)} & 1 & 0 & 0 & 0 & 0 & -u_1 x_w^{(1)} & -u_1 y_w^{(1)} & -u_1 z_w^{(1)} & -u_1 \\ 0 & 0 & 0 & 0 & x_w^{(1)} & y_w^{(1)} & z_w^{(1)} & 1 & -v_1 x_w^{(1)} & -v_1 y_w^{(1)} & -v_1 z_w^{(1)} & -v_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_w^{(i)} & y_w^{(i)} & z_w^{(i)} & 1 & 0 & 0 & 0 & 0 & -u_i x_w^{(i)} & -u_i y_w^{(i)} & -u_i z_w^{(i)} & -u_i \\ 0 & 0 & 0 & 0 & x_w^{(i)} & y_w^{(i)} & z_w^{(i)} & 1 & -v_i x_w^{(i)} & -v_i y_w^{(i)} & -v_i z_w^{(i)} & -v_i \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_w^{(n)} & y_w^{(n)} & z_w^{(n)} & 1 & 0 & 0 & 0 & 0 & -u_n x_w^{(n)} & -u_n y_w^{(n)} & -u_n z_w^{(n)} & -u_n \\ 0 & 0 & 0 & 0 & x_w^{(n)} & y_w^{(n)} & z_w^{(n)} & 1 & -v_n x_w^{(n)} & -v_n y_w^{(n)} & -v_n z_w^{(n)} & -v_n \end{bmatrix}}_A = \underbrace{\begin{bmatrix} p_{11} \\ p_{12} \\ p_{13} \\ p_{14} \\ p_{21} \\ p_{22} \\ p_{23} \\ p_{24} \\ p_{31} \\ p_{32} \\ p_{33} \\ p_{34} \end{bmatrix}}_p = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Known
Unknown

Camera Calibration Procedure

Step 5: Solve for p

$$A\mathbf{p} = 0$$

Let's look into the property of matrix P first ...

Property of Projection Matrices: Scale

We know that:

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} \equiv k \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} \quad (k \neq 0 \text{ is any constant})$$

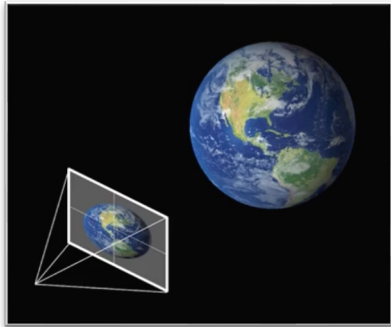
That is:

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix} \equiv k \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

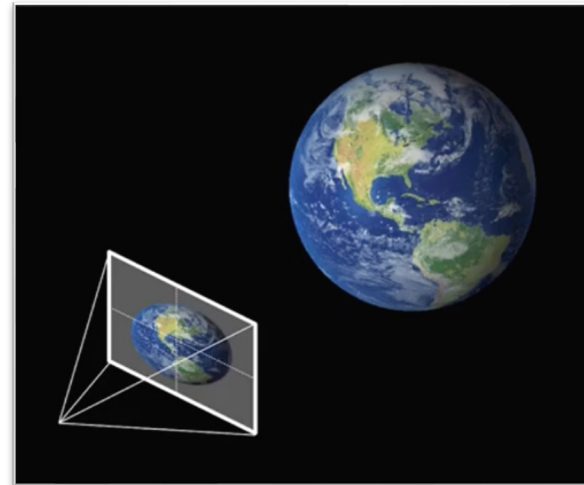
Therefore, Projection Matrices P and kP produce the same homogenous pixel coordinates.

Projection Matrix P is defined only up to a scale.

Property of Projection Matrices: Scale



Scale = k_1



Scale = k_2

Scaling projection matrix, implies simultaneously scaling the world and camera, which does not change the image

Solve P

Option 1: Set scale so that: $p_{34} = 1$

Option 2: Set scale so that: $\|p\|^2 = 1$

We want $A\mathbf{p}$ as close to 0 as possible and $\|\mathbf{p}\|^2 = 1$:

$$\min_p \|A\mathbf{p}\|^2 \quad \text{such that } \|\mathbf{p}\|^2 = 1$$

It can be proven that eigenvector $\mathbf{p}$ with **smallest eigenvalue** λ of matrix $A^T A$ is the solution.

Finally rearrange solution $\mathbf{p}$ to form the projection matrix P .

Decompose Projection Matrices to Intrinsic and Extrinsic Matrices

We know that:

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} = \underbrace{\begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{M_{int}} \underbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{M_{ext}}$$

Given that K is an **Upper Right Triangular** matrix and R is an **Orthonormal** matrix, it is possible to uniquely “decouple” K and R from their product using “**QR factorization**”.

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x \\ 0 & f_y & o_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = KR$$

Decompose Projection Matrices to Intrinsic and Extrinsic Matrices

We know that:

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

That is:

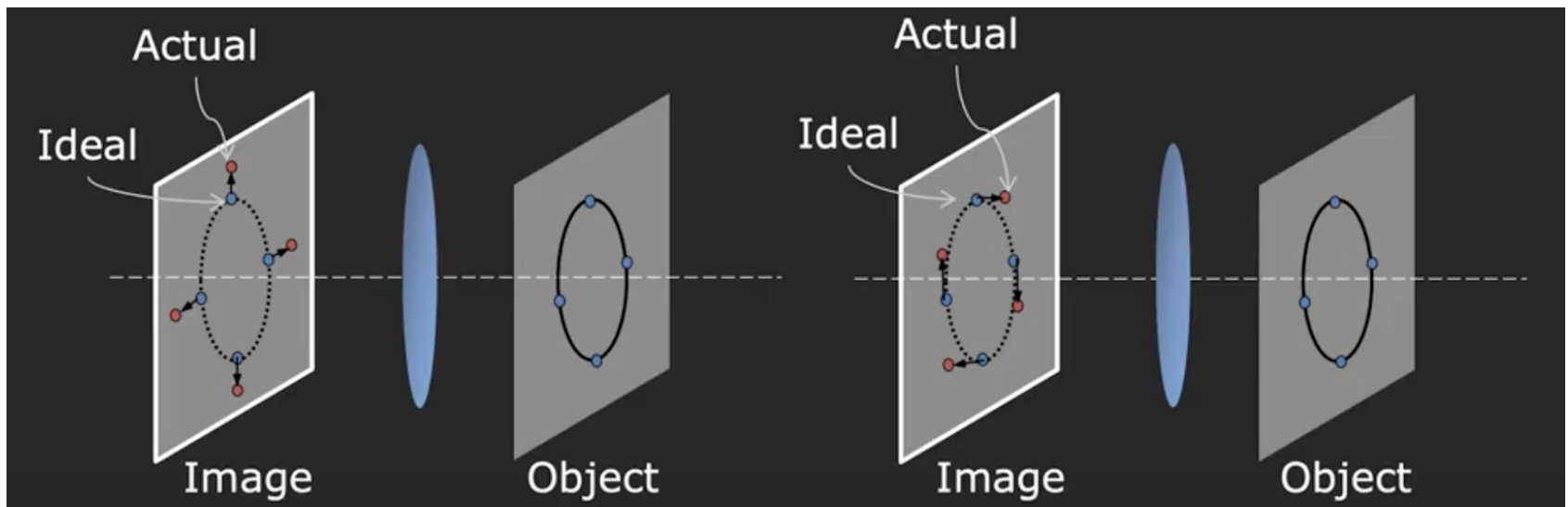
$$\begin{bmatrix} p_{14} \\ p_{24} \\ p_{34} \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x \\ 0 & f_y & o_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} = K\mathbf{t}$$

Therefore:

$$\mathbf{t} = \mathbf{K}^{-1} \begin{bmatrix} p_{14} \\ p_{24} \\ p_{34} \end{bmatrix}$$

Camera Distortions

Pinholes do not exhibit image distortions. But, lenses do!

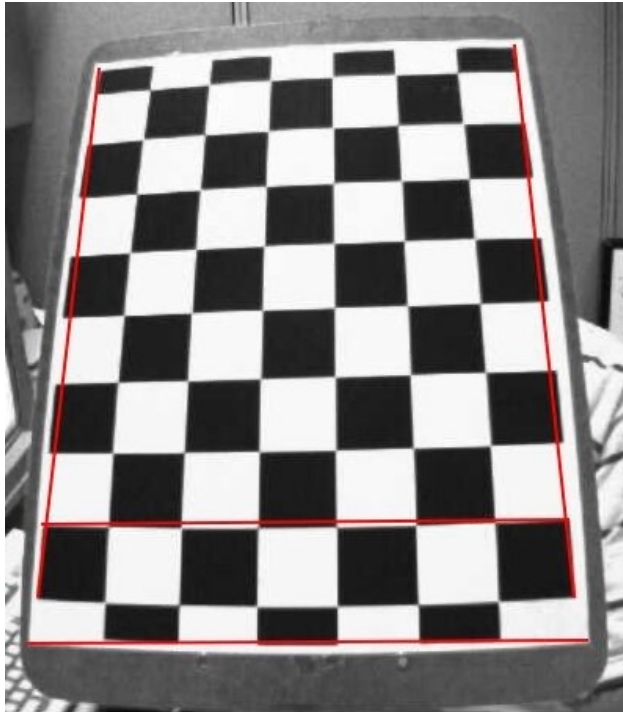


Radial Distortion

Tangential Distortion

The intrinsic model of the camera will need to include the distortion coefficients. We ignore distortions here.

Camera Distortions



Visual Localization Problem



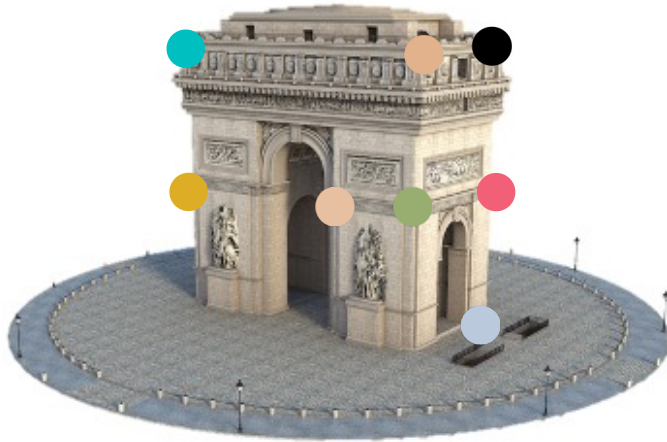
Known 3D Scene



Picture

Where was the picture taken?

Visual Localization Problem



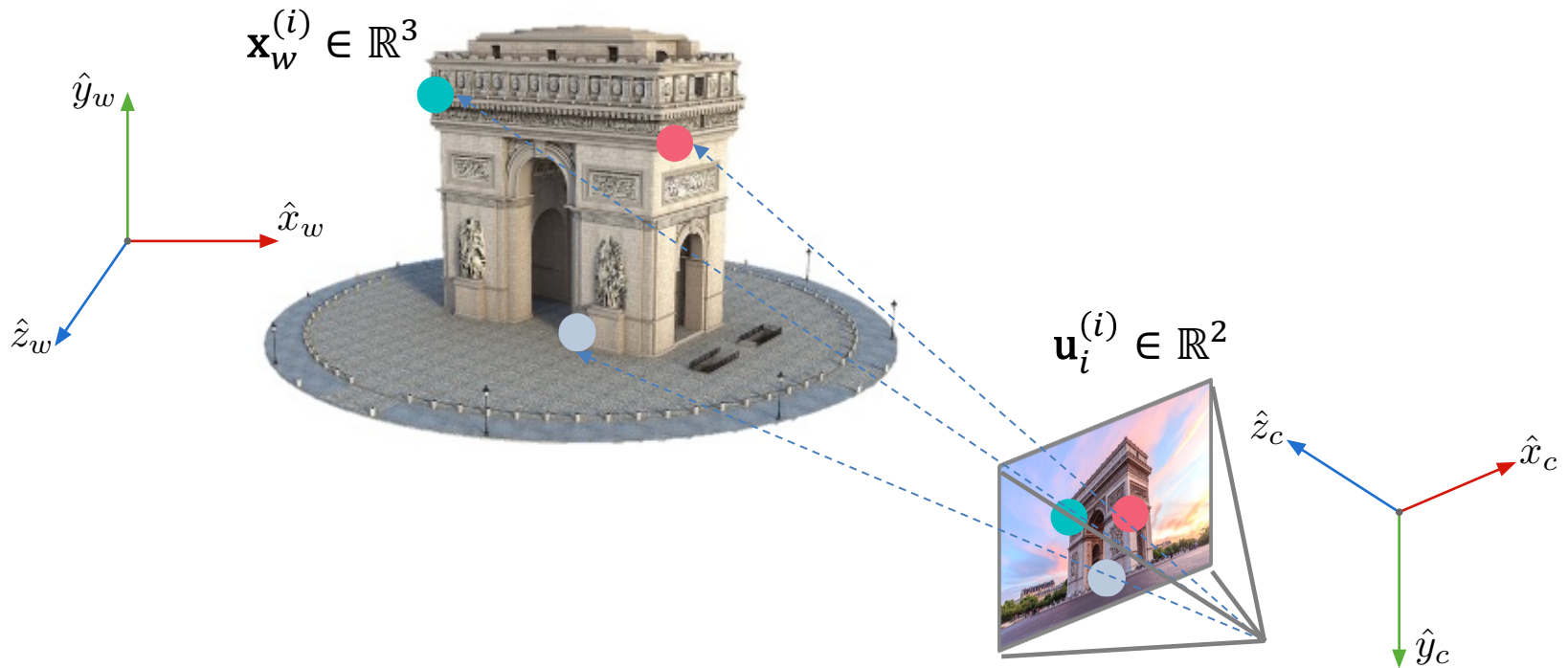
Known 3D Scene



Picture

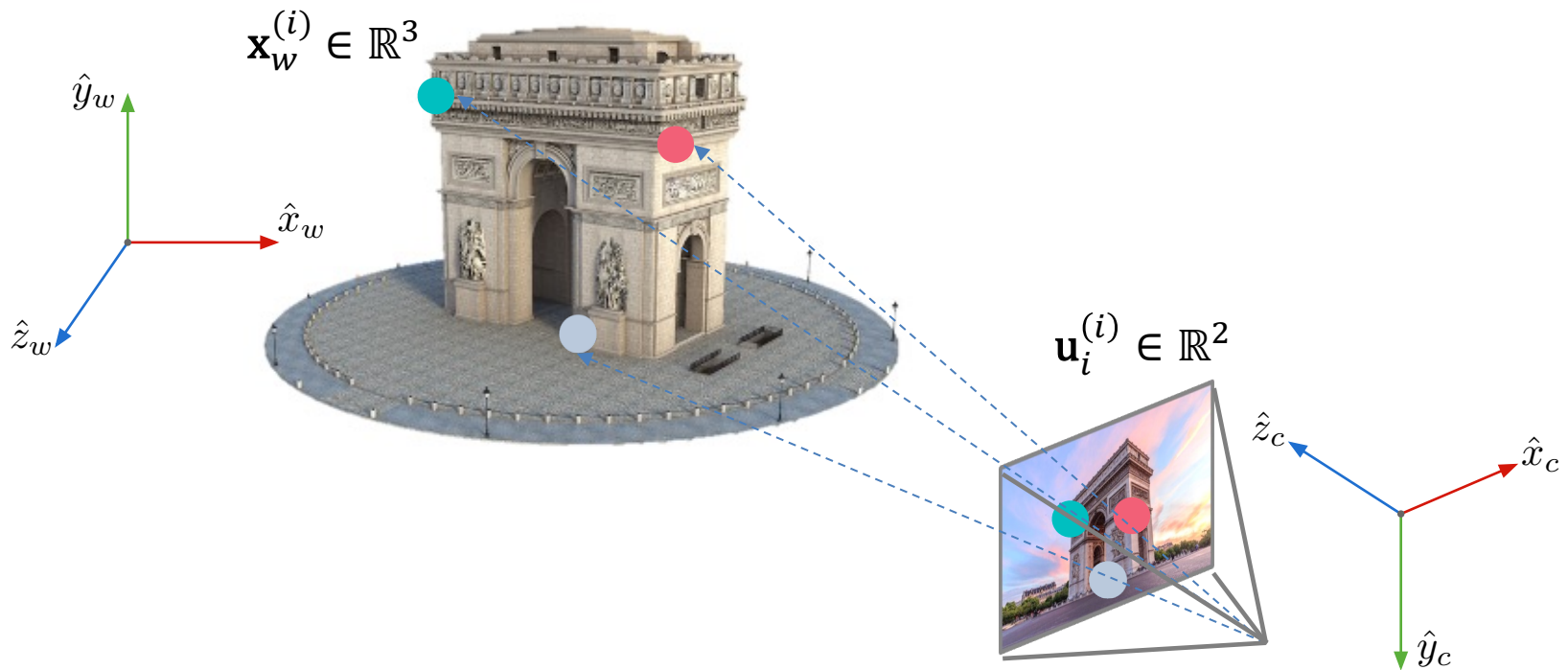
Step 1. Find 3D-2D correspondences

Visual Localization Problem



Step 2. Solve camera pose, given 3D-2D correspondences

Perspective-n-Point problem



Step 2. Solve camera pose, given 3D-2D correspondences

**This is called Perspective-n-Point (PnP) problem
if intrinsic parameters are known**

Perspective-n-Point problem

- How many unknown parameters?
 - 6 unknowns: 3 for rotation, 3 for translation
 - Usually called 6DoF pose estimation
- How many points do we need?

Direct Linear Transform (DLT)

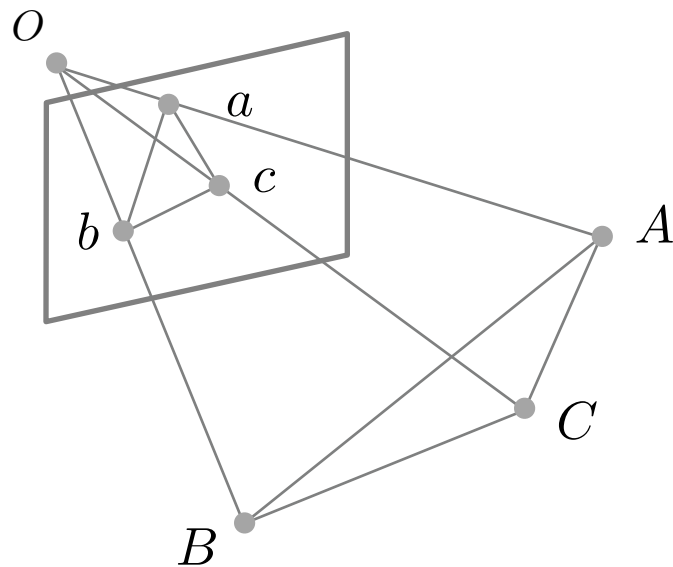
Recap: In camera calibration, we solve the problem with

$$\underbrace{\begin{bmatrix} u^{(i)} \\ v^{(i)} \\ 1 \end{bmatrix}}_{\text{Known}} \equiv \underbrace{\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix}}_{\text{Unknown}} \underbrace{\begin{bmatrix} x_w^{(i)} \\ y_w^{(i)} \\ z_w^{(i)} \\ 1 \end{bmatrix}}_{\text{Known}}$$

- This is also called a **direct linear transform (DLT) solution**.

P3P

- Solve the camera pose using the minimal number of points.



P3P

1. According to the cosine theorem,

$$OA^2 + OB^2 - 2OA \cdot OB \cdot \cos\langle a, b \rangle = AB^2$$

$$OB^2 + OC^2 - 2OB \cdot OC \cdot \cos\langle b, c \rangle = BC^2$$

$$OA^2 + OC^2 - 2OA \cdot OC \cdot \cos\langle a, c \rangle = AC^2$$

2. Divide by OC^2 , denote $x = \frac{OA}{OC}$, $y = \frac{OB}{OC}$,

$$x^2 + y^2 - 2xy \cos\langle a, b \rangle = AB^2/OC^2$$

$$y^2 + 1^2 - 2y \cos\langle b, c \rangle = BC^2/OC^2$$

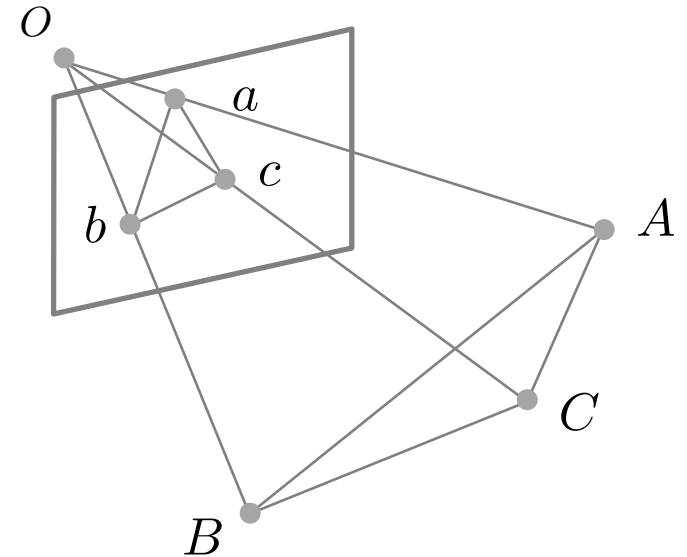
$$x^2 + 1^2 - 2x \cos\langle a, c \rangle = AC^2/OC^2$$

3. Denote $v = \frac{AB^2}{OC^2}$, $u = \frac{BC^2}{AB^2}$, $w = \frac{AC^2}{AB^2}$,

$$x^2 + y^2 - 2xy \cos\langle a, b \rangle - v = 0$$

$$y^2 + 1^2 - 2y \cos\langle b, c \rangle - uv = 0$$

$$x^2 + 1^2 - 2x \cos\langle a, c \rangle - wv = 0$$



P3P

1. According to the cosine theorem,

$$x^2 + y^2 - 2xy \cos\langle a, b \rangle - v = 0$$

2. Divide by OC^2 , denote $x = \frac{OA}{OC}$, $y = \frac{OB}{OC}$,

$$y^2 + 1^2 - 2y \cos\langle b, c \rangle - uv = 0$$

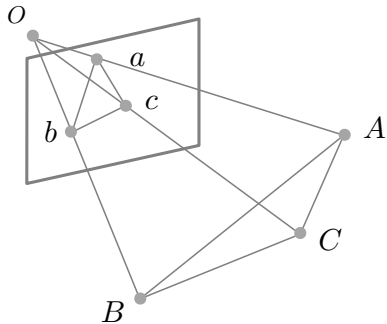
3. Denote $v = \frac{AB^2}{OC^2}$, $u = \frac{BC^2}{AB^2}$, $w = \frac{AC^2}{AB^2}$,

$$x^2 + 1^2 - 2x \cos\langle a, c \rangle - wv = 0$$

Use v to rearrange equation 3

$$(1 - u)y^2 - ux^2 - \cos\langle b, c \rangle y + 2uxy \cos\langle a, b \rangle + 1 = 0$$

$$(1 - w)x^2 - wy^2 - \cos\langle a, c \rangle x + 2wxy \cos\langle a, b \rangle + 1 = 0$$



Known: $\cos\langle a, b \rangle, \cos\langle b, c \rangle, \cos\langle a, c \rangle$
 u, w

Unknown: x, y



Binary
Quadratic
Equation

P3P

$$(1 - u)y^2 - ux^2 - \cos\langle b, c\rangle y + 2uxy \cos\langle a, b\rangle + 1 = 0$$

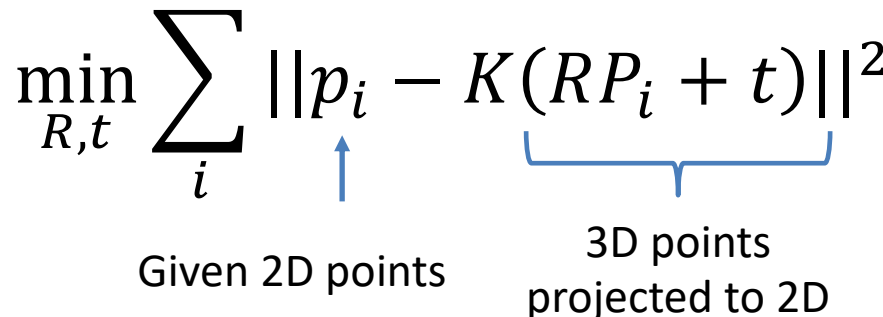
$$(1 - w)x^2 - wy^2 - \cos\langle a, c\rangle x + 2wxy \cos\langle a, b\rangle + 1 = 0$$

This binary quadratic equation has **4 possible answers**, we use **one extra point to determine** the most possible one.

PnP

- A more general solution for PnP problem:
 - minimizing the reprojection error

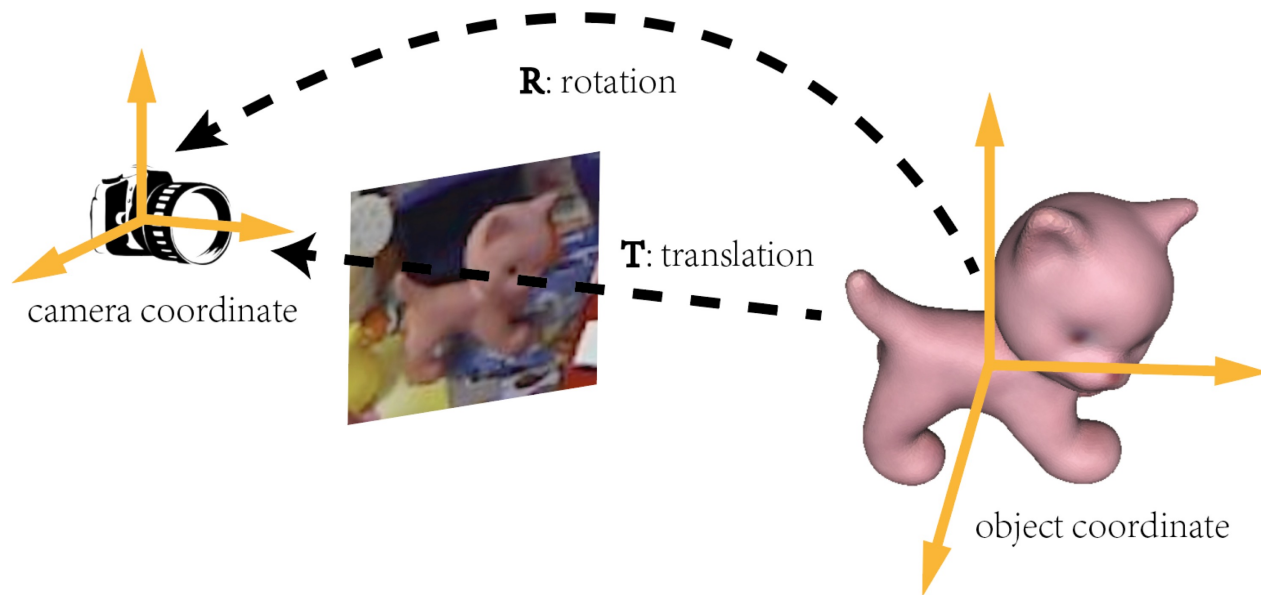
$$\min_{R,t} \sum_i ||p_i - \underbrace{K(RP_i + t)}_{\substack{\text{3D points} \\ \text{projected to 2D}}}||^2$$


Given 2D points 3D points
projected to 2D

- Initialized by P3P, optimized by Gauss-Newton

Object Pose Estimation

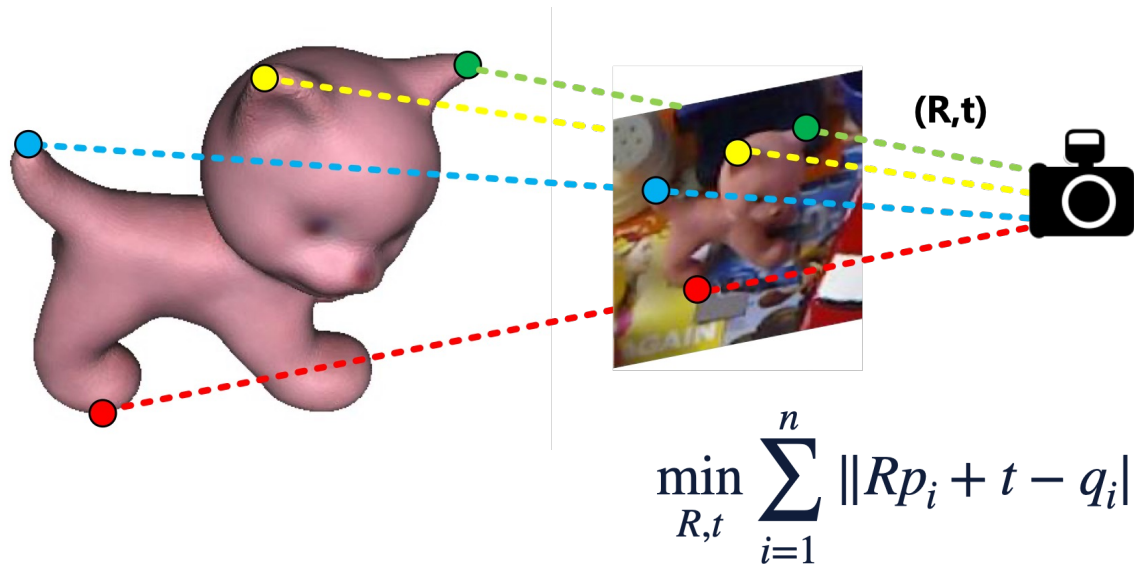
Estimate the **3D location and orientation** of an object relative to the camera coordinates



Object Pose Estimation

Similar to visual localization:

1. Find 3D-2D correspondences
2. Solve $\mathbf{R}$ and $\mathbf{t}$ by perspective-n-point (PnP) algorithm



Two remaining questions

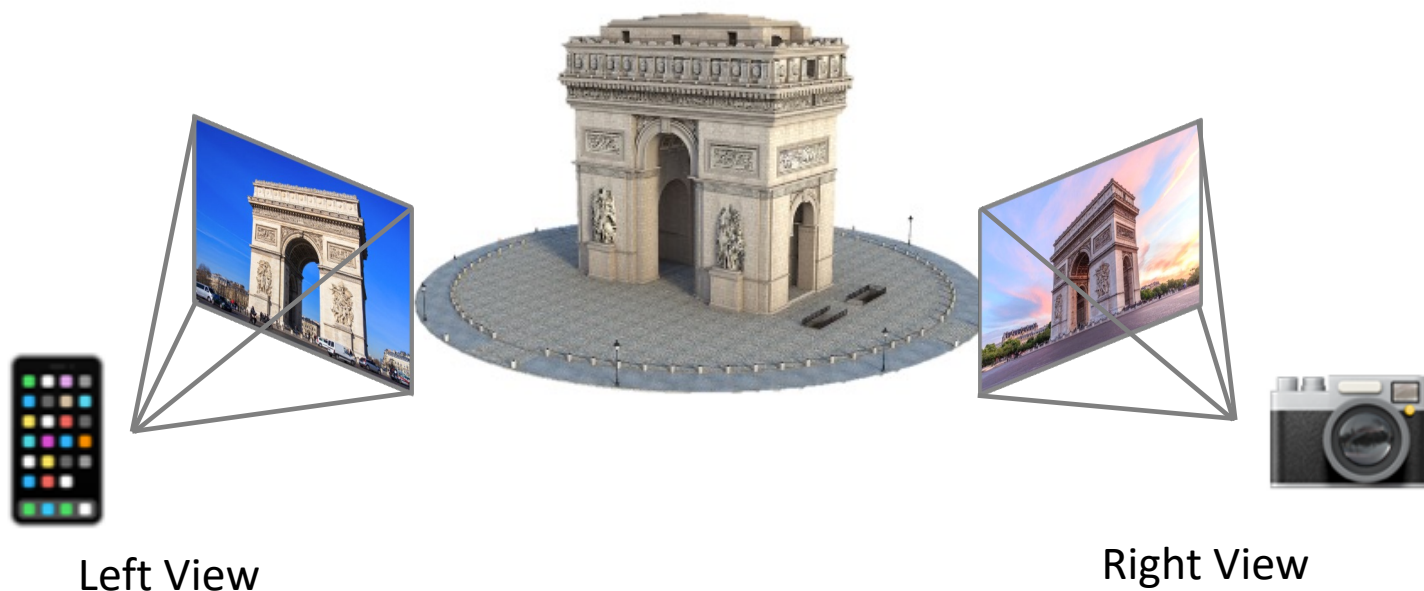
- How to reconstruct the 3D structure?
- How to find 3D-2D correspondences?

Today

1. Camera model
2. Camera calibration
- 3. Structure from motion**

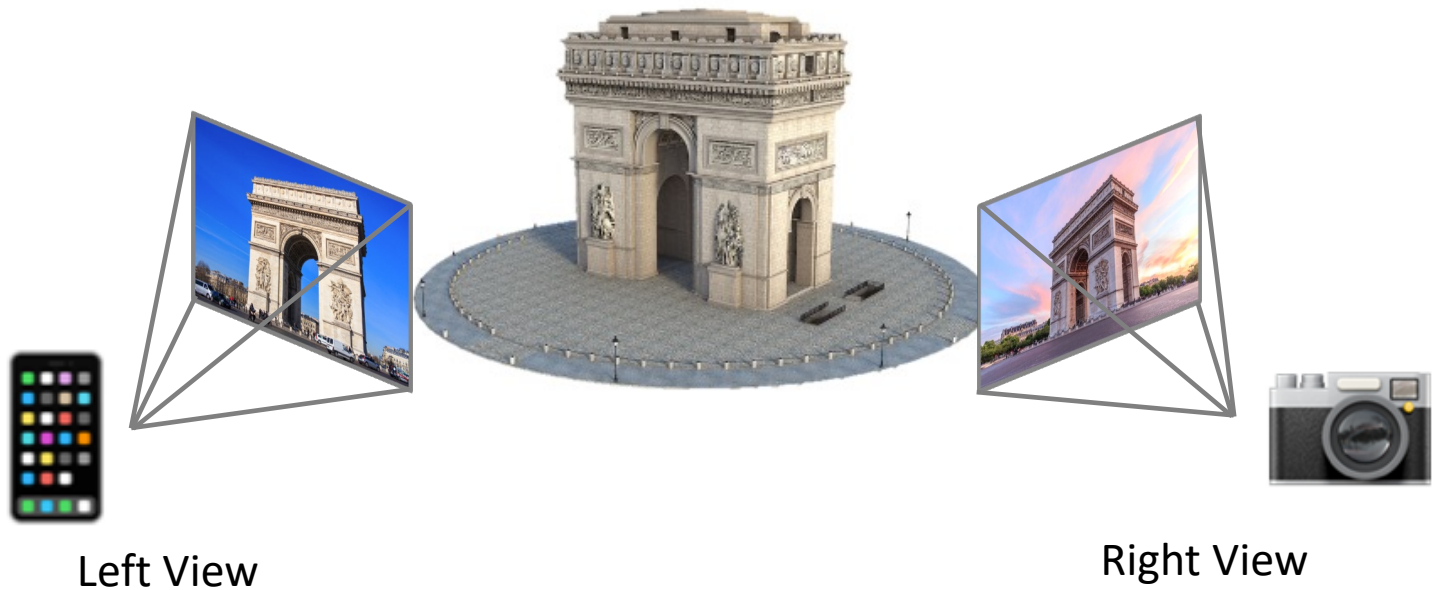
Structure from Motion (SfM)

Compute **3D structure** of the scene and **camera poses** from multiple views (images)



Intrinsics (f_x, f_y, o_x, o_y) are **known** for both views/cameras.
Extrinsics (relative position/orientation of cameras) are **unknown**.

Solving SfM



1. Assume intrinsic matrix K is known for each camera
2. Find a few reliable corresponding points
3. Find relative camera position $\mathbf{t}$ and orientation R
4. Find 3D position of scene points

Epipolar Geometry

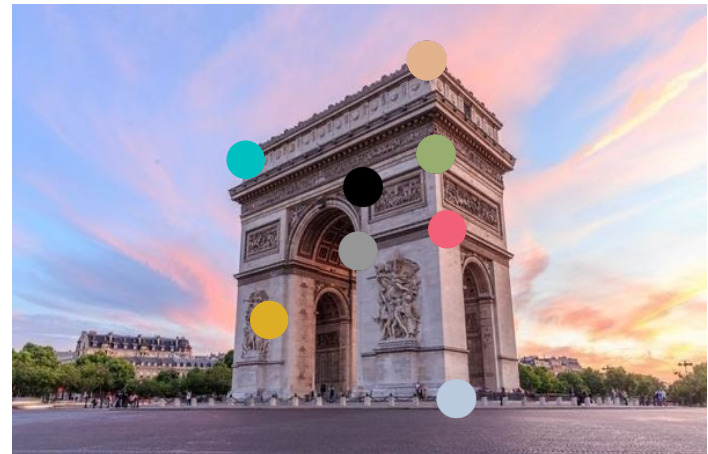
What is the geometric relation between the corresponding feature points?



● $(u_l^{(1)}, v_l^{(1)})$

⋮

● $(u_l^{(m)}, v_l^{(m)})$

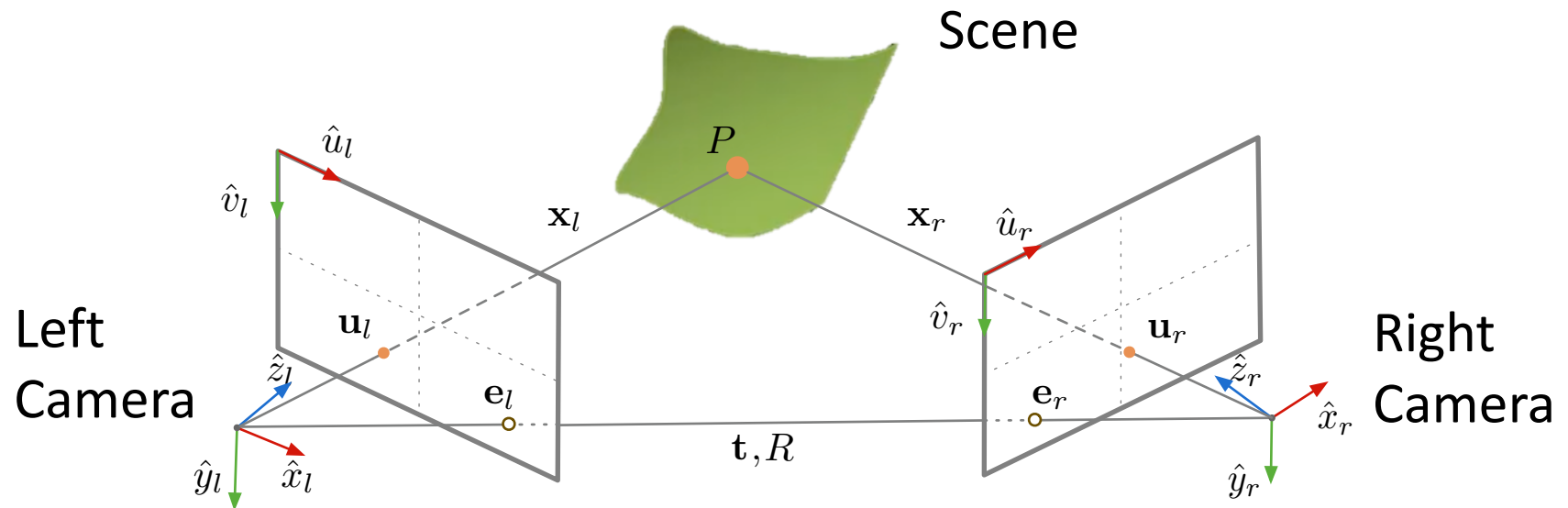


● $(u_r^{(1)}, v_r^{(1)})$

⋮

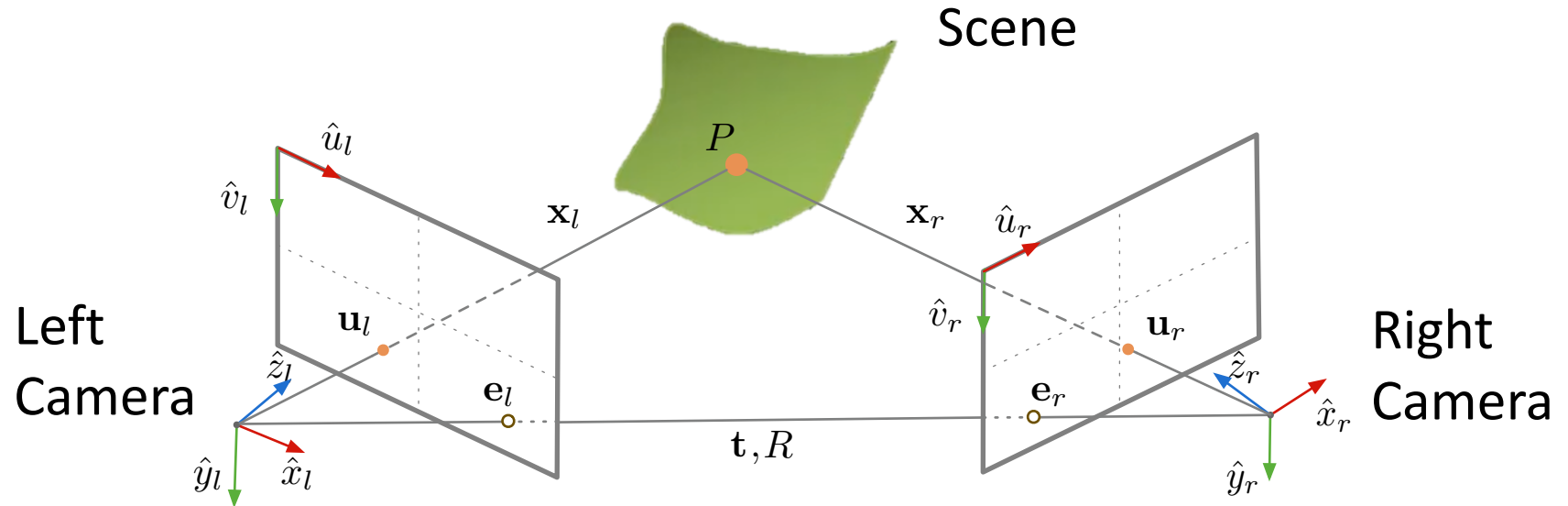
● $(u_r^{(m)}, v_r^{(m)})$

Epipolar Geometry



- Epipolar geometry describes the geometric relation between the 2D projections (u_l and u_r) of a 3D point in two views
- Epipolar geometry tells us how to solve $\mathbf{t}$ and R given a few pairs of 2D correspondences

Epipolar Geometry

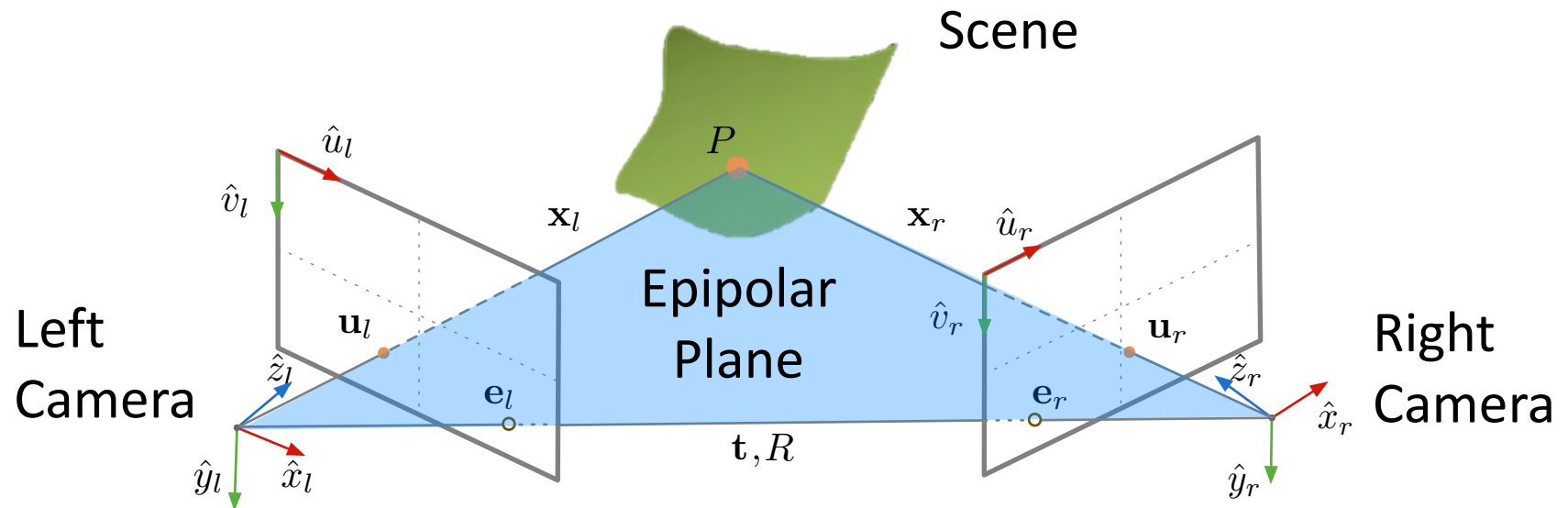


Epipole: Image point of origin/pinhole of one camera as viewed by the other camera.

$\mathbf{e}_l$ and $\mathbf{e}_r$ are the epipoles.

$\mathbf{e}_l$ and $\mathbf{e}_r$ are unique for a given pair.

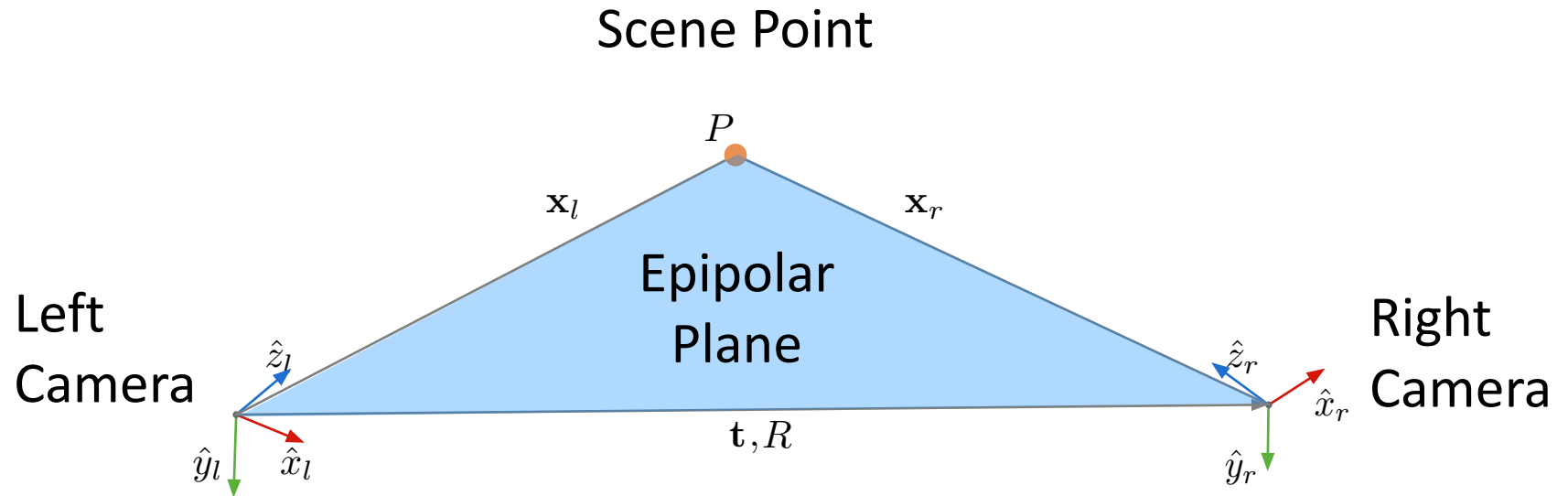
Epipolar Geometry



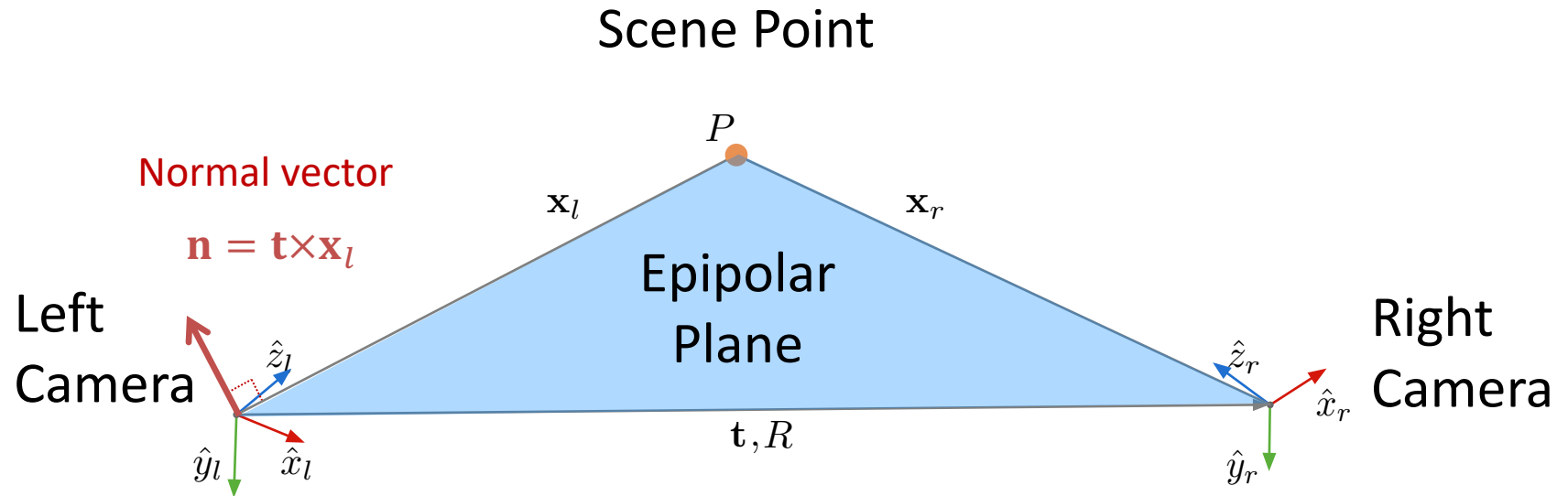
Epipolar Plane of Scene Point P : The plane formed by camera origins (O_l and O_r), epipoles (e_l and e_r) and scene point P .

Every scene point lies on a **unique epipolar plane**.

Epipolar Constraint



Epipolar Constraint



Normal vector to the epipolar plane: $\mathbf{n} = \mathbf{t} \times \mathbf{x}_l$

Dot product of $\mathbf{n}$ and $\mathbf{x}_l$ (perpendicular vectors) is zero:

$$\mathbf{x}_l \cdot (\mathbf{t} \times \mathbf{x}_l) = 0$$

Epipolar Constraint

Writing $\mathbf{x}_l \cdot (\mathbf{t} \times \mathbf{x}_l) = 0$ in matrix form:

$$[x_l \quad y_l \quad z_l] \begin{bmatrix} t_y z_l - t_z y_l \\ t_z x_l - t_x z_l \\ t_x y_l - t_y x_l \end{bmatrix} = 0$$

$$[x_l \quad y_l \quad z_l] \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix} = 0$$

$T_{\times}$

Matrix-vector form

We know that:

$$\mathbf{x}_l = R\mathbf{x}_r + \mathbf{t}$$

$\mathbf{t}_{3 \times 1}, R_{3 \times 3}$: Relative position and orientation between cameras

Epipolar Constraint

Substituting into the epipolar constraint gives:

$$\begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix} \left(\begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} + \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \right) = 0$$

$\mathbf{t} \times \mathbf{t} = 0$

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Essential Matrix E

$$E = T_{\times} R$$

Essential Matrix E: Decomposition

$$E = T_{\times} R$$

$$\begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} = \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

Given that $T_{\times}$ is a **Skew-Symmetric** matrix ($a_{ij} = a_{-ji}$) and R is an **Orthonormal** matrix, it is possible to “**decouple**” $T_{\times}$ and R from their product using “**Singular Value Decomposition**”.

Take Away: If E is known, we can calculate $\mathbf{t}$ and R .

How do we find E

Relates 3D position (x_l, y_l, z_l) of scene point w.r.t left camera to its 3D position (x_r, y_r, z_r) w.r.t. right camera

$$\mathbf{x}_l^T E \mathbf{x}_r = 0$$

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

3D position in left
camera coordinates

3x3 Essential Matrix

3D position in right
camera coordinates

Unfortunately, **we don't have $\mathbf{x}_l$ and $\mathbf{x}_r$.**

But we do know corresponding points in image coordinates.

Incorporating the Image Coordinates

Representing in matrix form:

$$\begin{aligned}u_l &= f_x^{(l)} \frac{x_l}{z_l} + o_x^{(l)} & v_l &= f_y^{(l)} \frac{y_l}{z_l} + o_y^{(l)} \\z_l u_l &= f_x^{(l)} x_l + z_l o_x^{(l)} & z_l v_l &= f_y^{(l)} y_l + z_l o_y^{(l)}\end{aligned}$$

Perspective projection equations for left camera:

$$z_l \begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} = \begin{bmatrix} z_l u_l \\ z_l v_l \\ z_l \end{bmatrix} = \begin{bmatrix} f_x^{(l)} x_l + z_l o_x^{(l)} \\ f_y^{(l)} y_l + z_l o_y^{(l)} \\ z_l \end{bmatrix} = \underbrace{\begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} \\ 0 & f_y^{(l)} & o_y^{(l)} \\ 0 & 0 & 1 \end{bmatrix}}_{\text{Known Camera Matrix } K_l} \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix}$$

Known
Camera Matrix K_l

Incorporating the Image Coordinates

Left camera

$$z_l \begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} \\ 0 & f_y^{(l)} & o_y^{(l)} \\ 0 & 0 & 1 \end{bmatrix}}_{K_l} \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix}$$

Right camera

$$z_r \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} \\ 0 & f_y^{(r)} & o_y^{(r)} \\ 0 & 0 & 1 \end{bmatrix}}_{K_r} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix}$$

$$\mathbf{x}_l^T = \begin{bmatrix} u_l & v_l & 1 \end{bmatrix} z_l K_l^{-1^T}$$

$$\mathbf{x}_r = K_r^{-1} z_r \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix}$$

Incorporating the Image Coordinates

Epipolar constraint:

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Rewriting in terms of image coordinates:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \cancel{z_l} K_l^{-1^T} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} K_r^{-1} \cancel{z_r} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

Depth of scene points doesn't affect the epipolar constraint!

Incorporating the Image Coordinates

Epipolar constraint:

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Rewriting in terms of image coordinates:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} K_l^{-1^T} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} K_r^{-1} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

Fundamental Matrix F

Epipolar constraint:

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Rewriting in terms of image coordinates:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

Fundamental Matrix F

$$E = K_l^T F K_r$$

The Tale of Missing Scale

Fundamental matrix acts on homogenous coordinates.

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0 = \begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} k f_{11} & k f_{12} & k f_{13} \\ k f_{21} & k f_{22} & k f_{23} \\ k f_{31} & k f_{32} & k f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

Fundamental Matrix F and kF describe the same epipolar geometry. That is, F is defined only up to a scale.

Usually adding a scale constraint on F

$$\|f\|^2 = 1$$

Relative Camera Pose Estimation

Find a set of **corresponding features (at least 8)** in left and right images (e.g. using SIFT or hand-picked).



$$\begin{aligned} & \text{cyan dot} \quad (u_l^{(1)}, v_l^{(1)}) \\ & \quad \quad \quad \vdots \\ & \text{red dot} \quad (u_l^{(m)}, v_l^{(m)}) \end{aligned}$$



$$\begin{aligned} & \text{cyan dot} \quad (u_r^{(1)}, v_r^{(1)}) \\ & \quad \quad \quad \vdots \\ & \text{red dot} \quad (u_r^{(m)}, v_r^{(m)}) \end{aligned}$$

Relative Camera Pose Estimation

Step A: For each correspondence i , write out epipolar constraint.

$$\underbrace{\begin{bmatrix} u_l^{(i)} & v_l^{(i)} & 1 \end{bmatrix}}_{\text{Known}} \underbrace{\begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}}_{\text{Unknown}} \underbrace{\begin{bmatrix} u_r^{(i)} \\ v_r^{(i)} \\ 1 \end{bmatrix}}_{\text{Known}} = 0$$

Expand the matrix to get linear equation:

$$\left(f_{11}u_r^{(i)} + f_{12}v_r^{(i)} + f_{13} \right) u_l^{(i)} + \left(f_{21}u_r^{(i)} + f_{22}v_r^{(i)} + f_{23} \right) v_l^{(i)} + f_{31}u_r^{(i)} + f_{32}v_r^{(i)} + f_{33} = 0$$

Relative Camera Pose Estimation

Step B: Rearrange terms to form a linear system.

$$\underbrace{\begin{bmatrix} u_l^{(1)} u_r^{(1)} & u_l^{(1)} v_r^{(1)} & u_l^{(1)} & v_l^{(1)} u_r^{(1)} & v_l^{(1)} v_r^{(1)} & v_l^{(1)} & u_r^{(1)} & v_r^{(1)} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_l^{(i)} u_r^{(i)} & u_l^{(i)} v_r^{(i)} & u_l^{(i)} & v_l^{(i)} u_r^{(i)} & v_l^{(i)} v_r^{(i)} & v_l^{(i)} & u_l^{(i)} & v_r^{(i)} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_l^{(m)} u_r^{(m)} & u_l^{(m)} v_r^{(m)} & u_l^{(m)} & v_l^{(m)} u_r^{(m)} & v_l^{(m)} v_r^{(m)} & v_l^{(m)} & u_l^{(m)} & v_r^{(m)} & 1 \end{bmatrix}}_{\substack{A \\ \text{Known}}} \underbrace{\begin{bmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ f_{22} \\ f_{23} \\ f_{31} \\ f_{32} \\ f_{33} \end{bmatrix}}_{\substack{f \\ \text{Unknown}}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Af = 0$$

Relative Camera Pose Estimation

Step C: Find least squares solution for fundamental matrix F .

We want $A\mathbf{f}$ as close to 0 as possible and $\|\mathbf{f}\|^2 = 1$:

$$\min_{\mathbf{f}} \|A\mathbf{f}\|^2 \text{ such that } \|\mathbf{f}\|^2 = 1$$

Constrained linear least squares problem

Like solving Projection Matrix during Camera Calibration.
Or, Homography Matrix for Image Stitching.

Rearrange solution $\mathbf{f}$ to form the fundamental matrix F .

Relative Camera Pose Estimation

Step D: Compute essential matrix E from known left and right intrinsic camera matrices and fundamental matrix F .

$$E = K_l^T F K_r$$

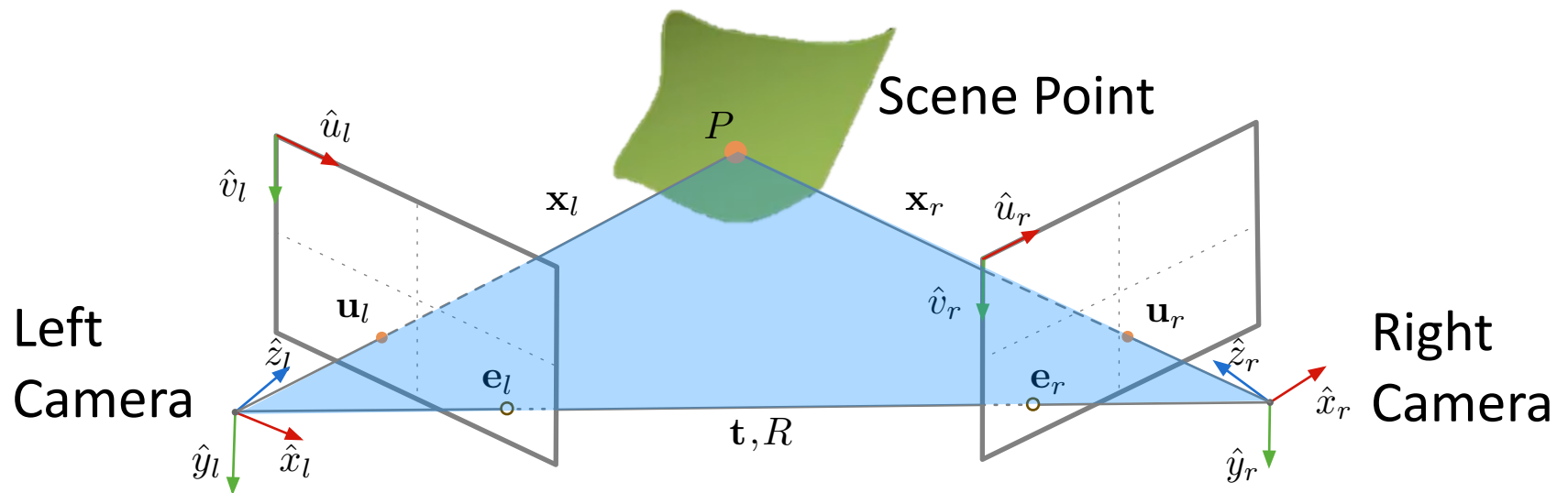
Step E: Extract R and $\mathbf{t}$ from E .

$$E = T_{\times} R$$

(Using Singular Value Decomposition)

Triangulation

Given corresponding 2D feature points and camera parameters,
 how to find the 3D coordinates of scene points?



$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix} \quad \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Triangulation

Left Camera Image Equation

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix}$$

Right Camera Image Equation

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

We also know the relative position and orientation between the two cameras.

$$\begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Triangulation

Left Camera Imaging Equation:

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

$$\tilde{\mathbf{u}}_l = P_l \tilde{\mathbf{x}}_r$$

Right Camera Imaging Equation:

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

$$\tilde{\mathbf{u}}_r = M_{int_r} \tilde{\mathbf{x}}_r$$

Triangulation

The imaging equations:

$$\tilde{\mathbf{u}}_r = M_r \tilde{\mathbf{x}}_r$$

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Known

Unknown

$$\tilde{\mathbf{u}}_l = P_l \tilde{\mathbf{x}}_r$$

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Known

Unknown

Rearranging the terms:

$$\begin{bmatrix} u_r m_{31} - m_{11} & u_r m_{32} - m_{12} & u_r m_{33} - m_{13} \\ v_r m_{31} - m_{21} & v_r m_{32} - m_{22} & v_r m_{33} - m_{23} \\ u_l p_{31} - p_{11} & u_l p_{32} - p_{12} & u_l p_{33} - p_{13} \\ v_l p_{31} - p_{21} & v_l p_{32} - p_{22} & v_l p_{33} - p_{23} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = \begin{bmatrix} m_{14} - m_{34} \\ m_{24} - m_{34} \\ p_{14} - p_{34} \\ p_{24} - p_{34} \end{bmatrix}$$

Triangulation

$$\begin{bmatrix} u_r m_{31} - m_{11} & u_r m_{32} - m_{12} & u_r m_{33} - m_{13} \\ v_r m_{31} - m_{21} & v_r m_{32} - m_{22} & v_r m_{33} - m_{23} \\ u_l p_{31} - p_{11} & u_l p_{32} - p_{12} & u_l p_{33} - p_{13} \\ v_l p_{31} - p_{21} & v_l p_{32} - p_{22} & v_l p_{33} - p_{23} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = \begin{bmatrix} m_{14} - m_{34} \\ m_{24} - m_{34} \\ p_{14} - p_{34} \\ p_{24} - p_{34} \end{bmatrix}$$

$A_{4 \times 3}$
Known

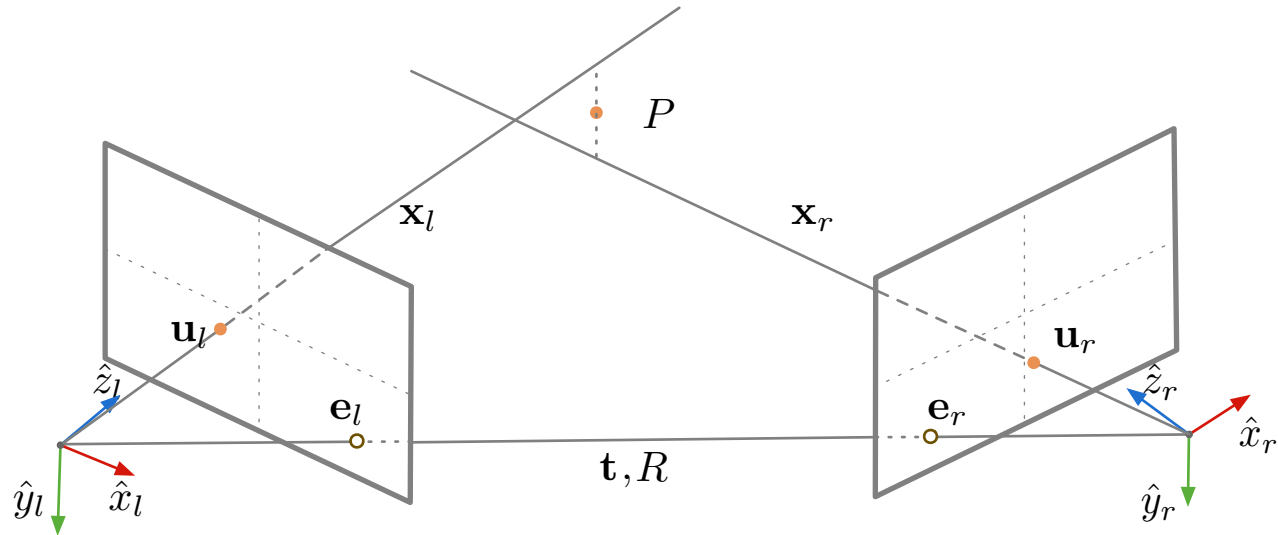
$\mathbf{x}_r$
Unknown

$\mathbf{b}_{4 \times 1}$
Known

Find **least squares solution** by:

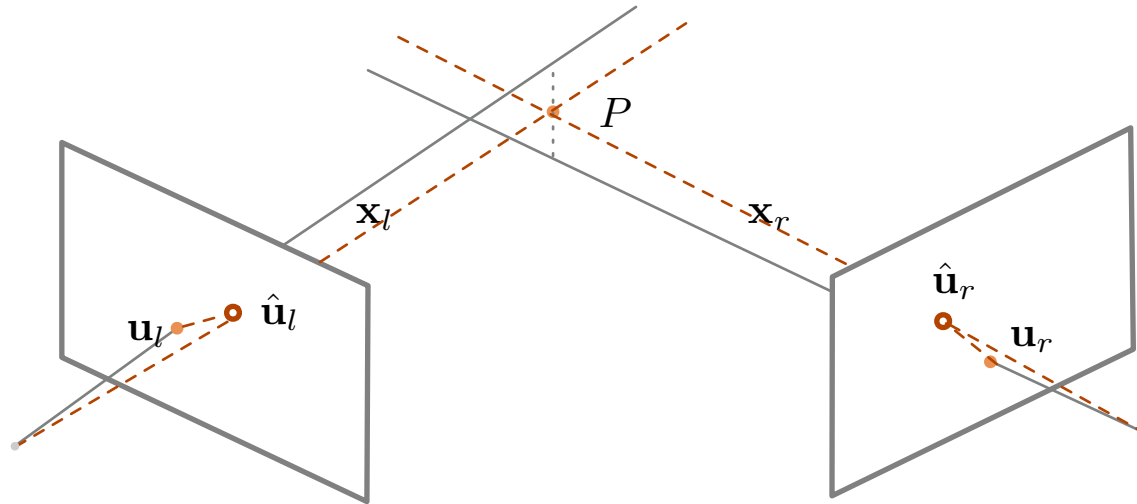
$$\mathbf{x}_r = (A^T A)^{-1} A^T \mathbf{b}$$

Triangulation: nonlinear solution



Generally, rays $\mathbf{x}_l$ and $\mathbf{x}_r$ will not exactly intersect.

Triangulation: nonlinear solution

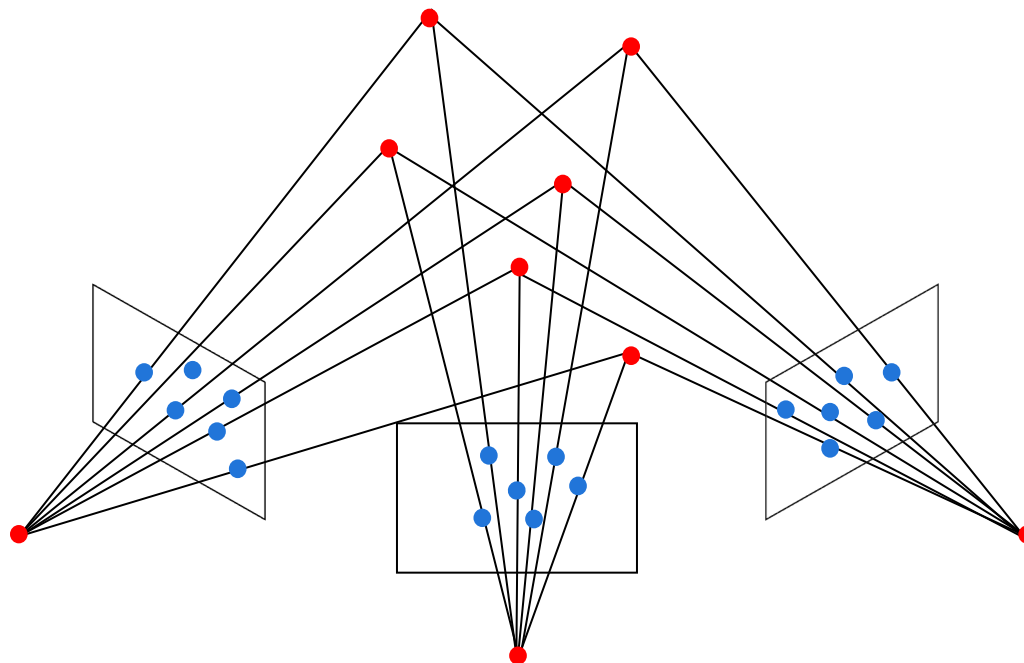


Minimize **reprojection error**:

$$cost(\mathbf{P}) = ||\mathbf{u}_l - \hat{\mathbf{u}}_l||^2 + ||\mathbf{u}_r - \hat{\mathbf{u}}_r||^2$$

Can we also optimize $R, \mathbf{t}$?

Multi-frame Structure from Motion

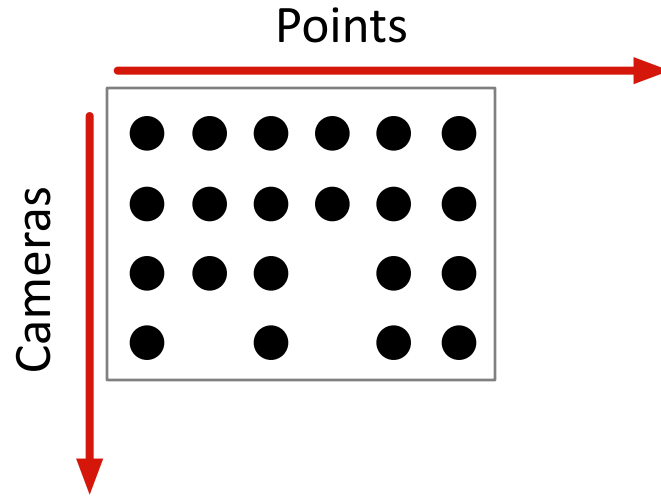


Given m images of n 3D points

$$\mathbf{u}_j^{(i)} = P_{proj}^{(i)} \mathbf{P}_j \quad (i = 1, \dots, m \text{ and } j = 1, \dots, n)$$

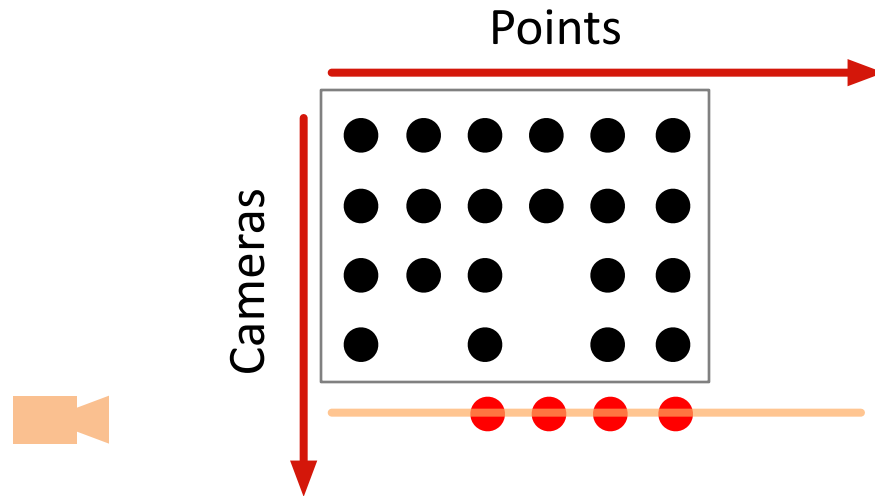
Estimate m projection matrices $P_{proj}^{(i)}$ and n 3D points $\mathbf{P}_j$ from the mn corresponding 2D points $\mathbf{u}_j^{(i)}$

Sequential Structure from Motion



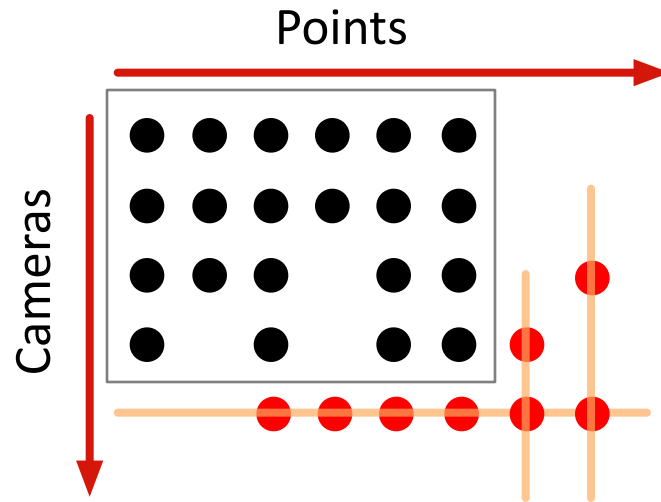
1. Initialize camera motion and scene structure

Sequential Structure from Motion



1. Initialize camera motion and scene structure
2. For each additional view
 - Determine projection matrix of **new camera** using all the known 3D points that are visible in its image

Sequential Structure from Motion

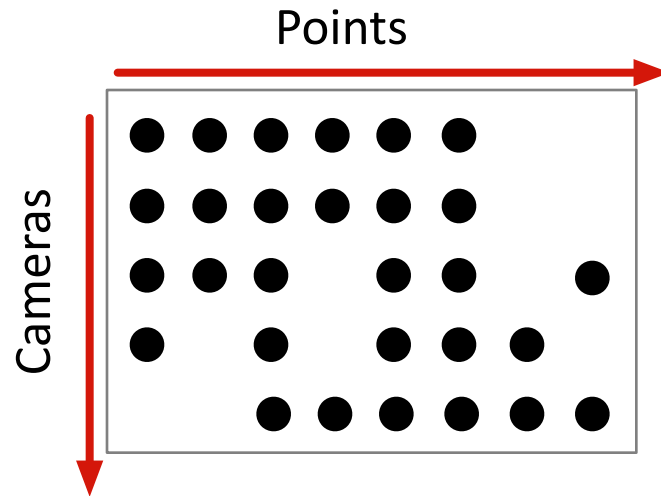


1. Initialize camera motion and scene structure

2. For each additional view

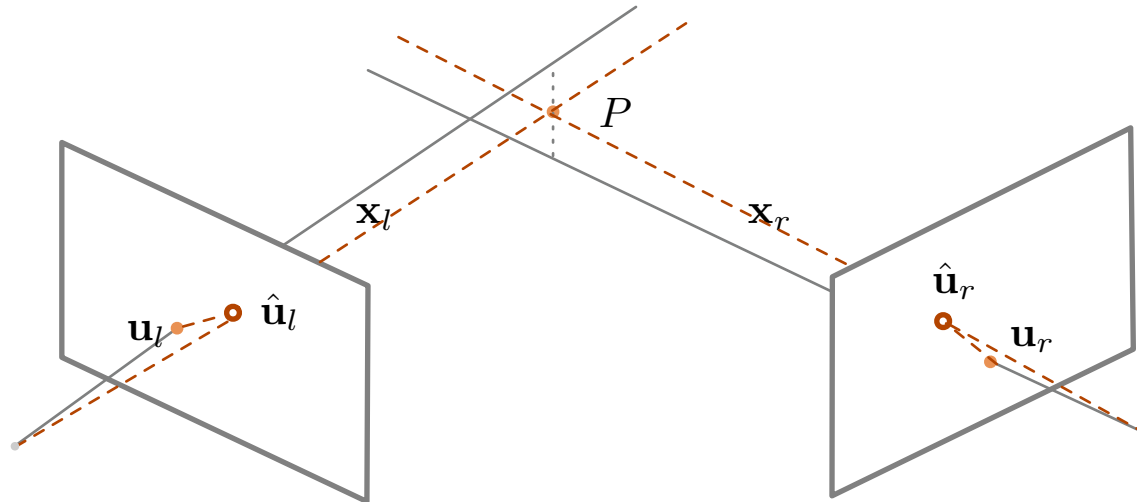
- Determine projection matrix of new camera using all the known 3D points that are visible in its image
- Refine and extend structure: compute **new 3D points**, re-optimize existing points that are also seen by this camera

Sequential Structure from Motion



1. Initialize camera motion and scene structure
2. For each additional view
 - Determine projection matrix of new camera using all the known 3D points that are visible in its image
 - Refine and extend structure: compute new 3D points, re-optimize existing points that are also seen by this camera
3. Refine structure and motion: Bundle Adjustment

Recap: Non-linear Solution

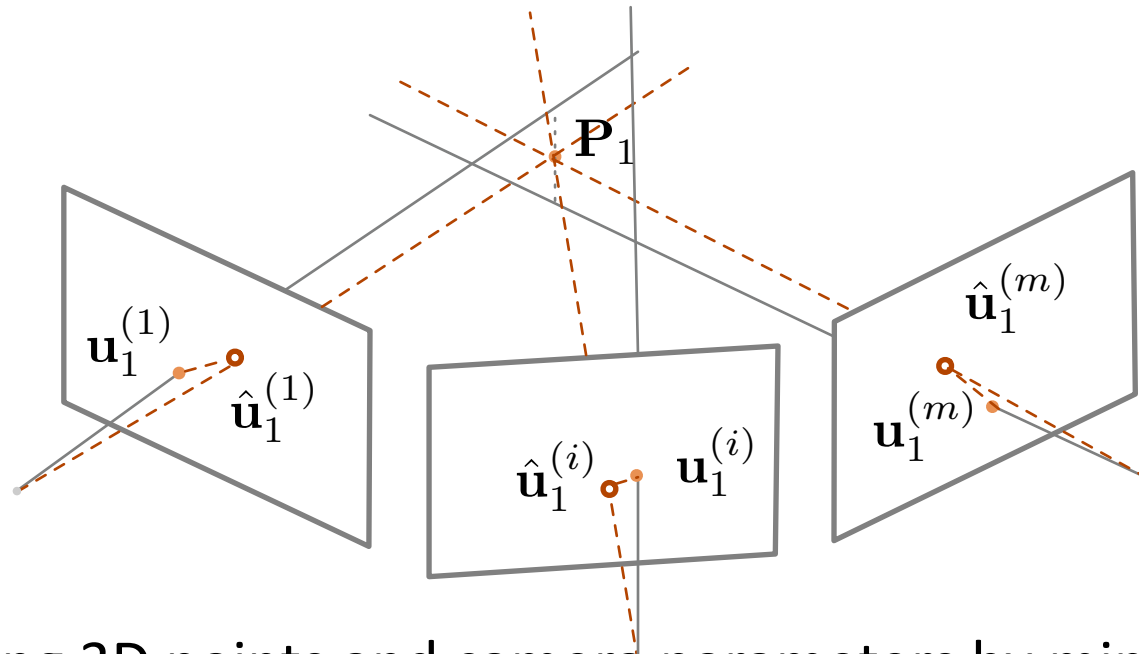


Minimize **reprojection error**:

$$cost(\mathbf{P}) = \|\mathbf{u}_l - \hat{\mathbf{u}}_l\|^2 + \|\mathbf{u}_r - \hat{\mathbf{u}}_r\|^2$$

Can we also optimize $R, \mathbf{t}$?

Bundle adjustment



Refining 3D points and camera parameters by minimizing **reprojection error** over all frames:

$$E(P_{proj}, \mathbf{P}) = \sum_{i=1}^m \sum_{j=1}^n \|u_j^{(i)} - P_{proj}^{(i)} \mathbf{P}_j\|^2$$

(Using Levenberg-Marquardt Algorithm, e.g. Ceres-Solver from Google)

COLMAP



Sparse model of central Rome using 21K photos produced by COLMAP's SfM pipeline.

COLMAP is a general-purpose **Structure-from-Motion (SfM)** and **Multi-View Stereo (MVS)** pipeline with a graphical and command-line interface. It offers a wide range of features for reconstruction of **ordered and unordered image collections**.

Homepage: colmap.github.io

Paper: Structure-from-motion revisited, Schonberger J L, Frahm J M. , CVPR 2016.

COLMAP



Dense models of several landmarks produced by COLMAP's MVS pipeline.

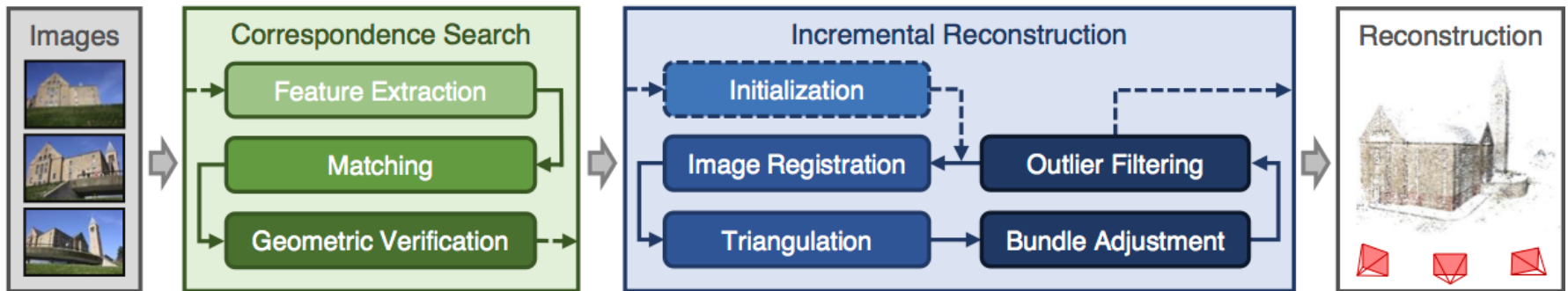
will be covered in the next lecture...

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Incremental SfM pipeline



- Question?